

NEW FROM SOLAX

X1-MINI G4



X1-MINI G4

0.6kW / 0.7kW / 0.8kW
1.1kW / 1.5kW / 2.0kW
2.5kW / 3.0kW / 3.3kW

Features

Upgraded Performance

- DC Input: 200% oversizing, 16A current, ultra wide MPPT voltage range
- In-built global MPP scan for higher yields

Optimized Monitoring

- 10s level interval of data update(Optional, new WiFi dongle required)
- Multiple interfaces for presenting data

Advanced Safety

- AC/DC built-in Type II SPD (Optional) and I-V curve diagnosis supported
- Support external Rapid Shutdown device (RSD)
- Optional ARC detection (AFCI)
- Built-in export control function

Flexible Adaptability

- Adapt to Home EV Charger Solution
- Mass management and broad extension via Datahub
- Extendable parallel solution via Modbus supported

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X1-MINI-0.6K-G4 X1-MINI-0.7K-G4 X1-MINI-0.8K-G4 X1-MINI-1.1K-G4 X1-MINI-1.5K-G4 X1-MINI-2.0K-G4 X1-MINI-2.5K-G4 X1-MINI-3.0K-G4 X1-MINI-3.3K-G4

DC INPUT

Max. PV input power [Wp]	1200	1400	1600	2200	3000	4000	5000	6000	6600
Max. PV input voltage [V]	450	450	450	450	450	450	550	550	550
Start-up voltage [V]	50	50	50	50	50	50	50	50	50
Nominal input voltage [V]	360	360	360	360	360	360	360	360	360
MPPT voltage range [V]	40~450	40~450	40~450	40~450	40~450	40~450	40~550	40~550	40~550
No. of MPP trackers/Strings per MPP tracker					1/1				
Max. PV input current [A]					16				
Max. input short circuit current Isc [A]					22				

AC OUTPUT

Rated AC output power [W]	600	700	800	1100	1500	2000	2500	3000	3300
Rated AC output current [A]	2.6	3.1	3.5	4.8	6.5	8.7	10.9	13.1	14.4
Max. AC output apparent power [VA]	600	770	800	1210	1650	2200	2750	3300	3300
Max. AC output current [A]	3	3.5	3.7	5.5	7.5	10	12.5	15	15
Nominal AC voltage/AC voltage range [V]**	220/230/240;90~285				220/230/240;90~290				
Nominal AC frequency/AC frequency range [Hz]**					50/60;±5				
Power Factor range					0.8 leading~0.8 lagging				
THDi (Rated power) [%]					<3				

SYSTEM DATA

Max. efficiency [%]	98	98	98	98	98	98	98	98	98
European weighted efficiency [%]	96	96	96	96	97	97	97	97	97
Standby consumption (night) [W]					<1				
Ingress protection					IP66				
Operating ambient temperature range [°C]					-25~60				
Max. operating altitude [m]					<4000				
Relative humidity [%]					0~100				
Typical noise emission [dB]					25				
Dimensions (WxHxD) [mm]					290x206x120				
Net weight [kg]	5.2	5.2	5.2	5.2	5.2	5.2	5.5	5.5	
Cooling concept					Natural cooling				
Communication interfaces					USB/RS485/DRM, Optional: Meter/CT *	5.5			
Optional monitoring dongle					Pocket WiFi/LAN/4G				
Display					2 x LED + LCD(16 x 2) / APP				

PROTECTION

Over/under voltage protection					Yes				
DC isolation protection					Yes				
Monitoring ground fault protection					Yes				
Grid monitoring					Yes				
DC injection monitoring					Yes				
Back feed current monitoring					Yes				
Residual current detection					Yes				
Anti-islanding protection					Yes				
Over temperature protection					Yes				
SPD (DC/AC)					Type II / Type II (Optional)				
Arc-fault circuit interrupter (AFCI)					Optional				
AC auxiliary power supply (APS)					Optional				
DC Switch					Optional				

STANDARD

Safety					EN/IEC62109-1/2				
EMC					EN61000-6-1/2/3/4; EN61000-3-2/3/11/12				
Certification					IEC61727, EN50549, G98/G99, AS 4777.2, VDE4105, CEI 0-21, VFR				

* Not included in the package. Please purchase additionally.

** The AC voltage and the frequency range may vary from different country codes

* V1.5. Information may be subject to modify without notice.650.00030.00

NEW FROM SOLAX

X1-BOOST G4



X1-BOOST G4

2.5kW / 3.0kW / 3.3kW / 3.6kW
4.0kW / 4.2kW / 5.0kW / 6.0kW

Features

Superior Performance

- DC Input: 200% oversizing, 16A current, 50V startup voltage
- In-built global MPP scan for higher yield efficiency

User Friendly Monitoring

- 10s level interval of data update(Optional, new WiFi dongle required)
- Quick and easy configuration (WiFi&APP)

Assured Safety & Reliability

- AC/DC built-in Type II SPD (Optional) and I-V curve diagnosis supported
- Optional external RSD (Rapid Shutdown) and integrated DC AFCI protection (Optional)
- In-built export control function

Superb Adaptability

- Home EV Charger and Heat Pump Solution adaptable
- Mass management and broad extension via Datahub
- Master/Slave parallel solution via Modbus supported

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DC INPUT

Max. PV input power [Wp]	6000	6000	6600	7200	8000	8000	10000	12000
Max. PV input voltage [V]	600	600	600	600	600	600	600	600
Start-up voltage [V]	50	50	50	50	50	50	50	50
Nominal input voltage [V]	360	360	360	360	360	360	360	360
MPPT voltage range [V]	40~560	40~560	40~560	40~560	40~560	40~560	40~560	40~560
No. of MPP trackers / String per MPP tracker	2 / 1							
Max. PV input current[A]	16 / 16							
Max. input short circuit current Isc [A]	22 / 22							

AC OUTPUT

Rated AC output power [W]	2500	3000	3300	3680	4000	4200	5000 ^①	6000
Rated AC output current [A]	10.9	13.1	14.4	16	17.4 ^⑥	18.3	21.7	26.1
Max. AC output apparent power [VA]	2750	3300	3630	4048 ^④	4000	4620	5000 ^②	6000
Max. AC output current [A]	12	14.4	15.8	17.6 ^⑤	17.4 ^⑥	20.1	21.7 ^③	27.3
Nominal AC voltage / AC voltage range [V] **	220/230/240;90~290							
Nominal AC frequency / AC frequency range [Hz] **	50/60;±5							
Power Factor range	0.8leading~0.8lagging							
THDi (Rated power) [%]	<3							

SYSTEM DATA

Max. Efficiency [%]	98							
European weighted efficiency [%]	97							
Standby consumption (night) [W]	<3							
Ingress protection	IP66							
Operating ambient temperature range [°C]	-25~60							
Max. operating altitude [m]	4000							
Relative humidity [%]	0~100							
Typical noise emission [dB]	25 ^⑦							
Dimensions (W x H x D) [mm]	404x274x146							
Net weight [kg]	11	11	11	11	11	11	11.5	11.5
Cooling concept	Natural cooling							
Communication interfaces	USB/RS485/DRM, Optional: Meter/CT *							
Optional monitoring dongle	Pocket WiFi/LAN/4G							
Display	2 x LED + LCD (16 x 2) / APP							

PROTECTION

Over / under voltage protection	Yes							
DC isolation protection	Yes							
Monitoring ground fault protection	Yes							
Grid monitoring	Yes							
DC injection monitoring	Yes							
Back feed current monitoring	Yes							
Residual current detection	Yes							
Anti-islanding protection	Yes							
Over temperature protection	Yes							
SPD (DC/AC)	Type II / Type II (Optional)							
Arc-fault circuit interrupter (AFCI)	Optional							
AC auxiliary power supply (APS)	Optional							
DC Switch	Optional							

STANDARD

Safety	IEC / EN 62109-1 / -2							
EMC	EN61000-6-1 / 2 / 3 / 4, EN61000-3-2 / 3 / 11 / 12							
Certification	IEC61727, EN50549, G98/G99, AS 4777.2, VDE4105, CEI 0-21, VFR, PPDS, TOR							

① 5000 (4600 for VDE4105, 4999 for AS4777.2) ② 5000 (4600 for VDE4105, 4999 for AS4777.2) ③ 21.7 (20 for VDE4105) ④ 4048 (3680 For G98/G99, TOR and PPDS) ⑤ 17.6 (16 For G98/G99, TOR and PPDS)

⑥ For models with internal fan(Optional), typical noise emission is 30 dB

* Not included in the package. Please purchase additionally, ** The AC voltage and the frequency range may vary from different country codes.

* V1.5. Information may be subject to modify without notice.650.00029.00

NEW FROM SOLAX

X1-SMART G2



X1-SMART G2

5KW / 6kW / 7kW
8kW / 9kW / 10kW

Features

Advanced performance

- 3 MPP trackers for multi-orientation installation
- Max. PV input current 20A compatible with advanced high-power panels
- 200% PV oversizing, and 110% AC overloading

User-friendly O&M

- Super convenient and quick plug-play communication terminal
- Aluminum AC wires supported
- 10s data refresh interval with new Cloud interface
- 24h consumption monitoring

Adaptable and Multifunctional

- Multiple units in parallel for larger system
- Global MPP scan to yield more
- Fast I-V curve scan and report for earlier panel issue diagnosis
- Home EV Charger and Heat Pump Solution adaptable

Robust safety protection

- Optional external RSD (Rapid Shut Down) and integrated DC AFCI protection (Optional)
- DC/AC both sides SPD2 protection (Optional modules to meet certification IEC/EN61643)
- Support 2030.5 and flexible control

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X1-SMART G2

	X1-SMT-5K-G2	X1-SMT-6K-G2	X1-SMT-7K-G2	X1-SMT-8K-G2	X1-SMT-9K-G2	X1-SMT-10K-G2
DC INPUT						
Max. PV input power [Wp]	10000	12000	14000	16000	18000	20000
Max. PV input voltage [V]	600					
Start-up voltage [V]	50					
Nominal input voltage [V]	360					
MPPT voltage range [V]	40~560					
No. of MPP trackers / Strings per MPP tracker	3/1					
Max. PV input current[A]	20					
Max. input short circuit current Isc [A]	25					
AC OUTPUT						
Rated AC output power [W]	4999	6000	7000	8000	9000	9999
Max. AC output apparent power [VA]	4999	6600	7700	8800	9900	9999
Rated AC output current [A]	21.8	26.1	30.5	34.8	39.2	43.5
Max. AC output current [A]	22.8	30	35	40	45	45.5
Nominal AC voltage/AC voltage range [V]	220/230/240; (90~290 V)					
Nominal AC frequency/AC frequency range [Hz]	50/60; ±5					
Power Factor range	0.8 leading ~ 0.8 lagging					
THDi (Rated power) [%]	<3					
SYSTEM DATA						
Max. Efficiency [%]	98.2					
European weighted efficiency [%]	97.5					
Standby consumption (night) [W]	<3					
Ingress protection	IP66					
Operating ambient temperature range [°C]	-25~60					
Max. operating altitude [m]	4000					
Relative humidity [%]	0~100					
Typical noise emission [dB]	<30					
Dimensions(WxHxD) [mm]	515×370×170					
Net weight [kg]	19.5					
Cooling concept	Natural cooling					
Communication interfaces	RS485/DRM/USB/CT/Meter*					
Optional monitoring dongle	WiFi/LAN/4G/WiFi+LAN/WiFi+4G					
PROTECTION						
Over/under voltage protection	Yes					
DC isolation protection	Yes					
Monitoring ground fault protection	Yes					
Grid monitoring	Yes					
DC injection monitoring	Yes					
Back feed current monitoring	Yes					
Residual current detection	Yes					
Anti-islanding protection	Yes					
Over temperature protection	Yes					
SPD(DC/AC)	Type II/Type II					
Arc-fault circuit interrupter(AFCI)	Optional					
AC auxiliary power supply(APS)	Optional					
DC Switch	Yes					
STANDARD						
Safety	IEC/EN 62109-1/-2					
EMC	EN61000-6-1/2/3/4, EN61000-3-2/3/11/12					
Certification	AS 4777.2:2020, G99, INMETRO, IEC61727					

NEW FROM SOLAX

X3-MIC-G2



X3-MIC-G2

3.0kW/4.0kW/5.0kW/6.0kW
8.0kW/10.0kW/12.0kW/15.0kW

Features

High-efficiency

- Maximum efficiency is up to 98.3%
- Low startup voltage, ultrawide MPPT voltage range
- 200% oversizing, 110% overloading output (Except 15kW model)
- In-built global MPP scan for higher yield efficiency

Safe

- IP66 protection
- Integrated SPD protection on both AC&DC

Smart

- Built-in export power control
- Remote setting and upgrading
- 24h monitoring and maintenance (Optional)
- Intelligent load management - heat pump (Adapter Box required)
- Multiple monitoring methods supported, Optional: WiFi/LAN/4G

Economic

- Ultra-high power density
- Max. MPP current per string is 16A, support high power solar panels

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DC INPUT

Max. PV input power [Wp]	6000	8000	10000	12000	16000	20000	24000	30000
Max. PV input voltage [V]	1000	1000	1000	1000	1000	1000	1000	1000
Start-up voltage [V]	150	150	150	150	150	150	150	150
Nominal input voltage [V]	640	640	640	640	640	640	640	640
MPPT voltage range [V]	120~980	120~980	120~980	120~980	120~980	120~980	120~980	120~980
No. of MPP trackers/Strings per MPP tracker	2(1/1)	2(1/1)	2(1/1)	2(1/1)	2(1/1)	2(1/1) ^①	2(2/1)	2(2/1)
Max. PV input current [A]	16/16	16/16	16/16	16/16	16/16	16/16 ^①	32/16	32/16
Max. input short circuit current Isc [A]	20/20	20/20	20/20	20/20	20/20	20/20 ^①	40/20	40/20

AC OUTPUT

Rated AC output power [W]	3000	4000	5000 ^②	6000	8000	10000 ^③	12000	15000 ^④
Rated AC output current [A]*	4.6/4.4	6.1/5.8	7.6/7.3	9.1/8.7	12.2/11.6	15.2/14.5	18.2/17.4	22.7/21.8
Max. AC output apparent power [VA]	3300	4400	5500 ^②	6600	8800	11000 ^③	13200	15000 ^④
Max. AC output current [A]	4.8	6.4	8.0	9.6	12.8	16.0	19.1	22.7
Nominal AC voltage/AC voltage range [V]**	220/380V, 230/400V, 3/(N)/PE; (95-285V)*							
Nominal AC frequency/AC frequency range [Hz]**	50/60; ±5							
Power Factor range	0.8 leading-0.8 lagging							
THDi (Rated power) [%]	<3							

SYSTEM DATA

Max. efficiency [%]	98.3							
European weighted efficiency [%]	97.8							
Standby consumption(night) [W]	<3							
Ingress protection	IP66							
Operating ambient temperature range [°C]	− 30 ~ + 60							
Max. operating altitude [m]	4000							
Relative humidity [%]	0~100							
Typical noise emission [dB]	<30	<30	<30	<30	<45	<45	<50	<50
Dimensions(WxHxD) [mm]	342*434*144.5				342*434*156			
Net weight [kg]	15.5	15.5	15.5	15.5	17	17	18	18
Cooling concept	Natural cooling				Smart fan cooling			
Communication interfaces	USB/RS485/DRM, Optional: Meter							
Optional monitoring dongle	Pocket WiFi / LAN / 4G							
Display	2 x LED + LCD(16 x 2) / APP							

PROTECTION

Over/under voltage protection	YES							
DC isolation protection	YES							
DC reverse-polarity protection	YES							
Grid monitoring	YES							
DC injection monitoring	YES							
Back feed current monitoring	YES							
Residual current detection	YES							
Anti-islanding protection	YES							
Over temperature protection	YES							
SPD (DC/AC)	Type II / Type II (Optional)							
Arc-fault circuit interrupter(AFCI)	Optional							
AC auxiliary power supply(APS)	Optional							

STANDARD

Safety	IEC/EN 62109-1; IEC/EN 62109-2; NB/T 32004							
EMC	IEC/EN 61000; NB/T 32004							
Certification	VDE4105, EN 50549, AS 4777.2, VDE4105, G98/G99, IEC 61727, IEC 62116, IEC 61683, IEC 60068, EN 50530, NB/T 32004							

① Input 1 is optional with two strings(Max. input current: 32A, Max. short circuit current: 40A)

② 4999.for.AS/NZS.4777.2 ③ 9999.for.AS/NZS.4777.2 ④ 14999.for.AS/NZS.4777.2

* The two data refer to different grid voltage 220V/230V

** The AC voltage and the frequency range may vary from different country codes

*V2.8. Information may be subject to modify without notice. 650.00003.00

NEW FROM SOLAX

X3-PRO G2



X3-PRO G2

8kW/10kW/12kW/15kW
17kW/20kW/25kW/30kW

Features

High-efficiency

- Maximum efficiency is up to 98.5%
- Low startup voltage, ultrawide MPPT voltage range
- 150% DC oversizing, 110% AC overloading output
- In-built global MPP scan for higher yield efficiency

Safe

- SPD type II protection on both AC&DC
- ARC protection (Optional)
- IP66 protection

Smart

- Built-in export power control
- Intelligent load management - heat pump (Adapter Box required)
- 24h monitoring and maintenance (Optional)
- Multiple monitoring methods supported, Optional: WiFi/LAN/4G

Economic

- Ultra-high power density
- Maximum 32A DC input current per MPP tracker, support high power solar panels
- Up to 3 MPPTs, 2 strings per MPPT
- Support Master/Slave parallel function

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DC INPUT								
Max. PV array input power [Wp]	12000	15000	18000	22500	25500	30000	37500	45000
Max. PV input voltage [V]	1100	1100	1100	1100	1100	1100	1100	1100
Start startup voltage [V]	200	200	200	200	200	200	200	200
Nominal input voltage [V]	650	650	650	650	650	650	650	650
MPP tracker voltage range [V]	160~980							
No. of MPP trackers	2	2	2	2	2	2	3	3
Strings per MPP tracker	2	2	2	2	2	2	2	2
Max. PV input current [A]	32/32	32/32	32/32	32/32	32/32	32/32	32/32/32	32/32/32
Isc PV Array Short Circuit current [A]	40/40	40/40	40/40	40/40	40/40	40/40	40/40/40	40/40/40
AC OUTPUT								
Rated AC output power [kW]	8000	10000 ^①	12000	15000	17000	20000	25000	30000 ^②
Rated AC output current [A]*	12.2/11.6	15.2/14.5	18.2/17.4	22.8/21.8	25.8/24.7	30.3/29	37.9/36.3	45.5/43.5
Max. AC output apparent power [VA]	8800	11000 ^③	13200	16500	18700	22000	27500	30000 ^③
Max. AC output current [A]	13.2	16	19.3	24.2	27.5	33.6	41.8	45.5
Nominal AC voltage/AC voltage range [V]**	220/380V, 230/400V, 3/N/PE, 3/PE; 95-285V							
Nominal AC frequency/AC frequency range [Hz]**	50/60; ±5							
Power Factor range	0.8 leading ~ 0.8 lagging							
THDi (Rated power) [%]	<3							
SYSTEM DATA								
Max. efficiency [%]	98.20	98.20	98.20	98.30	98.30	98.30	98.50	98.50
Euro efficiency [%]	97.70	97.70	97.70	97.80	97.80	97.80	98.00	98.00
Standby consumption (Night) [W]	<3							
Ingress protection	IP66							
Operating ambient temperature range [°C]	-30~+60 (Derating above 45)							
Max. operation altitude [m]	4000 (Derating above 3000)							
Relative humidity [%]	0~100							
Typical noise emission [dB]	<35	<35	<35	<55	<55	<55	<55	<58
Storage temperature [°C]	-30~+60							
Dimensions (WxHxD) [mm]	482x417x181							
Weight [kg]	24.5			26			28	
Cooling concept	Natural cooling			Smart fan cooling				
Communication interfaces	USB / RS485 / DRM, Optional: Meter							
Optional monitoring dongle	Pocket WiFi/LAN/4G							
Display	2 x LED + LCD (16 x 2) / APP							
PROTECTION								
Over/under voltage protection	YES							
DC isolation protection	YES							
Grid monitoring	YES							
DC injection monitoring	YES							
Residual current detection	YES							
Anti-islanding protection	YES							
Over Temp protection	YES							
SPD (DC/AC)	Type II / Type II							
AC auxiliary power supply (APS)	Optional							
Arc-fault circuit interrupter (AFCI)	Optional							
STANDARD								
Safety	IEC/EN 62109-1; IEC/EN 62109-2; NB/T 32004							
EMC	IEC/EN 61000; NB/T 32004							
Certification	VDE4105; EN 50549; AS 4777.2; VDE4105; IEC 61727; IEC 62116; IEC 61683; IEC 60068; EN 50530; NB/T 32004							

* The two data refer to different grid voltage 220V/230V

** The AC voltage and the frequency range may vary from different country codes

①9999 for AS4777.2 ②29999 for AS4777.2

*V2.5. Information may be subject to modify without notice. 650.00004.00

NEW FROM SOLAX

X3-MEGA G2



X3-MEGA G2
40kW/50kW/60kW

Features

More energy harvest

- Maximum efficiency 98.4%
- 180~1000Vdc MPPT voltage range
- Maximum 6 MPPTs, 2 strings per MPP tracker
- 150% PV oversizing input, 110% overloading output
- Maximum 32A MPPT current

Safety & Reliability

- IP66 protection level
- AFCI protection (Optional)
- Both AC&DC SPDs(Type II) inside, Type I+II SPD is optional

Intelligence for easy maintenance and economy

- Built-in export power control
- Remote setting and upgrading
- Smart I-V Curve Diagnosis supported
- Aluminium AC cable connection available
- Fuse free design with smart string current monitoring
- Night-time reactive power compensation
- 24 hours operation monitoring (Optional)
- Smart air cooling technique results in long lifetime of fans
- Advanced heat dissipation technology makes the system more than 10% lighter and smaller

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X3-MEGA G2

X3-MGA-40K-G2

X3-MGA-50K-G2

X3-MGA-60K-G2

DC INPUT

Max. PV array input power [kWp]	60	75	90
Max. PV input voltage [V]		1100	
Startup voltage [V]		200	
Nominal input voltage [V]		600	
MPP tracker voltage range [V]		180~1000	
No. of MPP trackers	4	5	6
Strings per MPP tracker	2	2	2
Max. PV input current per MPPT [A]		32	
Isc PV Array Short Circuit current per MPPT [A]		46	

AC OUTPUT

Rated AC output power [kW]	40	50	60
Rated AC output current [A]*	60.6 / 58	75.8 / 72.5	90.9 / 87
Max. AC output apparent power [kVA]	44	55	66
Max. AC output current [A]*	66.7 / 63.8	83.3 / 79.7	100 / 95.7
Nominal AC voltage [V]	220/380V, 230/400V, 3/N/PE, 3/PE		
AC voltage range [V]**	304~460		
Nominal AC frequency / AC frequency range [Hz]**	50/60; ±5		
Power Factor range	0.8 leading ~ 0.8 lagging		
THDi (Rated power) [%]	<3		

SYSTEM DATA

Max. efficiency [%]	98.4		
Euro. efficiency [%]	98.1		
Standby consumption [W] @Night	<2		
Ingress protection	IP66		
Operating ambient temperature range [°C]	-30~+60		
Max. operation altitude [m]	4000 (Derating above 3000)		
Relative humidity [%]	0~100		
Dimensions [WxHxD] [mm]	630*521*286		
Weight [kg]	44	44.5	45.5
Cooling concept	Smart fan cooling		
Communication interfaces	RS485 / USB / DRM / PLC(Optional)		
Optional monitoring dongle	Pocket WiFi / LAN / 4G		
Display	LCD (16x2, optional) / LEDx4		

PROTECTION

Over/under voltage protection	YES
Over current protection	YES
DC isolation protection	YES
Grid monitoring	YES
DC injection monitoring	YES
Residual current detection	YES
Anti-islanding protection	YES
String fault detection	YES
Over temperature protection	YES
SPD (DC/AC)	Type II / Type II
Arc-fault circuit interrupter (AFCI)	Optional
AC auxiliary power supply (APS)	Optional

STANDARD

Safety	IEC/EN 62109-1; IEC/EN 62109-2; NB/T 32004
EMC	EN/IEC 61000; NB/T 32004
Certification	VDE4105; EN 50549; AS 4777.2; VDE4105; IEC 61727; IEC 62116; IEC 61683; IEC 60068; EN 50530; NB/T 32004

* The two data refer to different grid voltage 220V/230V

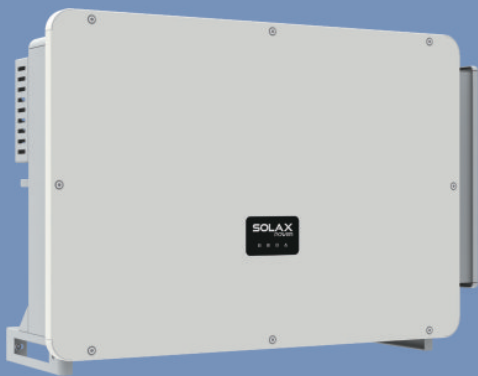
** The AC voltage and the frequency range may vary from different country codes

*V2.7. Information may be subject to modify without notice. 650.00002.00



PARTNER

NEW FROM SOLAX X3-FORTH



X3-FORTH

75kW/80kW/100kW/110kW
120kW/125kW/136kW/150kW

Features

More energy harvest

- Maximum efficiency up to 99%
- 180~1000Vdc MPPT voltage range
- Maximum 12 MPPTs, 2 strings per MPP tracker
- 150% PV oversizing input, 110% overloading output
- Maximum 32A MPPT current

Safety & Reliability

- IP66 protection level
- AFCI protection (Optional)
- AC terminal temperature detection
- Both AC&DC SPDs(Type II) inside, Type I+II SPD is optional

Intelligence for easy maintenance and economy

- Built-in export power control
- Remote setting and upgrading
- 24 hours operation monitoring
- Smart I-V Curve Diagnosis supported
- Night-time reactive power compensation
- Aluminium AC cable connection available
- Power line communication (PLC)(Optional)
- Fuse free design with smart string current monitoring
- Smart air cooling technique results in long lifetime of fans
- Advanced heat dissipation technology makes the system more than 5% lighter and smaller

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X3-FORTH (THREE PHASE)

X3-FTH-75K X3-FTH-80K X3-FTH-100K X3-FTH-110K X3-FTH-120K X3-FTH-125K X3-FTH-136K-MV X3-FTH-150K-MV

DC INPUT

Max. PV input power [kWp]	120	120	150	165	180	188	204	225
Max. PV input voltage [V] ^①	1100							
Nominal input voltage [V] ^②	580/600	580/600	580/600	580/600	580/600	580/600	730/785	730/785
Start-up voltage [V]	200							
MPPT voltage range [V] ^③	180-1000							
No. of MPP trackers	9	9	9/12 ^④	9/12 ^④	12	12	12	12
Strings per MPP tracker	2							
Max. PV input current per MPPT [A]	32							
Max. input short circuit current Isc per MPPT [A]	46							

AC OUTPUT

Rated AC output power [kW]	75	80	100	110	120	125	136	150
Rated AC output current [A]②	113.7/108.7	121.3/116	151.6/145	166.7/159.5	181.9/174	189.4/181.2	157.1/145.4	173.2/160.4
Max. AC output apparent power [kVA]	75	88	110	121kW	132kW	132	149.6kW	165kW
Max. AC output current [A]②	113.7/108.7	133.4/127.6	166.7/159.5	183.4/175.4	200/191.3	200/191.3	172.8/160	190.6/176.5
Nominal AC voltage [V]	220/380, 230/400, 3W/(N)/PE						500/540, 3W/PE	
AC voltage range [V]⑤	304 ~ 480						425 ~ 594	
Nominal AC frequency/AC frequency range [Hz]⑤	50/60; ±5							
THDi (Rated power) [%]	<3							
Adjustable power factor range	0.8 leading ~ 0.8 lagging							

SYSTEM DATA

Max. Efficiency [%]	98.6	98.6	98.6	98.6	98.6	98.6	99.0	99.0
European weighted efficiency [%]	98.3						98.5	
Ingress protection	IP66							
Operating ambient temperature range [°C]	-25 ~ +60							
Max. operation altitude [m]	4000							
Relative humidity [%]	0 ~100							
Dimensions (WxHxD) [mm]	985*660*327.5							
Weight [kg]	83	83	83	83	87	87	87	87
Cooling concept	Smart fan cooling							
Communication interfaces	RS485 / USB / DRM / PLC(Optional)							
Optional monitoring dongle	Pocket WiFi/LAN/4G							
Display	LCD(16x2, optional)/LEDx4							

PROTECTION

Over/under voltage protection	YES
DC isolation protection	YES
Grid monitoring	YES
DC injection monitoring	YES
Residual current detection	YES
Anti-islanding protection	YES
String fault detection	YES
SPD (DC/AC)	Type II / Type II
Arc-fault circuit interrupter(AFCI)	Optional
AC terminals over temperature detection	YES
AC auxiliary power supply(APS)	Optional
Power line communication(PLC)	Optional

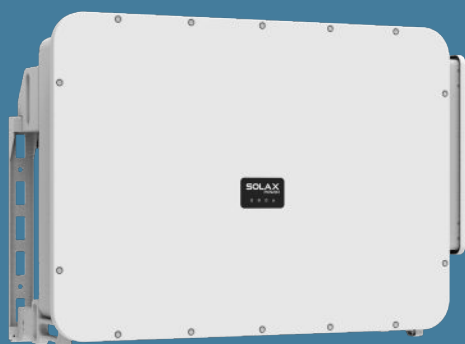
STANDARD

Safety	IEC/EN 62109-1; IEC/EN 62109-2; NB/T 32004
EMC	IEC/EN 61000; NB/T 32004
Certification	EN 50549; AS4777.2; VDE4105; IEC 61727; IEC 62116; IEC 61683; IEC 60068; EN 50530; NB/T 32004; NRS097; G99; RD1699; PPDS; CE10-21; CE10-16; VFR 2019; VDE4110

① The maximum input voltage represents the highest DC voltage threshold for the inverter, approaching which power derating occurs and any further increase might lead to potential damage to the inverter.
 ② The two data refer to different grid voltage 220V/230V(75~125kW models) or 500V/540V(136~150kW models).
 ③ Please refer to the user manual for the MPPT voltage range at full load condition. PV voltage out of the full load MPPT range will trigger the inverter derating protection or possibly cause damage to the inverter if it exceeds the Max. input voltage.
 ④ 9/12MPPTs is optional for 100kW and 110kW models.
 ⑤ The AC voltage and the frequency range may vary from different country codes.

NEW FROM SOLAX

X3-GRAND HV



X3-GRAND HV
300kW / 320kW
333kW / 350kW

Features

More energy harvest

- Maximum efficiency up to 99%
- 460~1500 Vdc MPPT voltage range
- Maximum 6 MPPTs, 5 strings per MPP tracker
- Maximum 75A MPPT current

Safety & Reliability

- IP66 protection level and NEMA4X for US
- AFCI protection (Optional)
- AC terminal temperature detection
- Both AC&DC SPDs(Type II) inside, Type I+II SPD is optional

Intelligence for easy maintenance and economy

- Built-in export power control
- Remote setting and upgrading
- 24 hours operation monitoring
- Power line communication(PLC)(Optional)

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DC Input

Max. PV input voltage	1500V
Nominal input voltage	1080V
Start-up voltage	500V
MPPT voltage range	460~1500V
Max. PV input current per MPPT	75A
Max. input short circuit current Isc per MPPT	115A
No. of MPP trackers	6
Strings per MPP tracker	5

AC Output

Rated AC output power	300kVA	320kVA	333kVA	352kVA
Rated AC output current	216.6A	231A	240A	254A
Max. AC output apparent power	330kVA	352kVA	333kVA	352kVA
Max. AC output current	238.2A	254A	254A	254A
Nominal AC voltage	3W+PE,800V	3W+PE,800V	3W+PE,800V	3W+PE,800V
AC voltage range	640-920V	640-920V	640-920V	640-920V
Nominal AC frequency	50Hz/60Hz	50Hz/60Hz	50Hz/60Hz	50Hz/60Hz
Power Factor range	>0.99 (0.8 leading-0.8 lagging)	>0.99 (0.8 leading-0.8 lagging)	>0.99 (0.8 leading-0.8 lagging)	>0.99 (0.8 leading-0.8 lagging)
THDi (Rated power)	<1%	<1%	<1%	<1%

EFFICIENCY

Max. efficiency	99%
European weighted efficiency	98.80%

GENERAL

Dimensions (WxHxD)	1145.5*806*405mm
Net weight	<120kg
Topology	Transformerless
Standby consumption (night)	<15W
Operating ambient temperature range	-30 ~ +60°C
Relative humidity	0-100%
Ingress protection	IP66 / NEMA4X
Cooling concept	Smart fan-cooling
Max. operation altitude	4000m
DC connection	MC4 (Max.6 mm2)
AC connection	OT/DT terminal(Max.400 mm2)
Display	LED, LCD(Optional), APP
Communication	RS485/USB/(Optional: WiFi/LAN/4G/PLC)/DI/Sunspec/2030.5/Modbus_RTU/Modbus_TCP

PROTECTION

DC switch	Yes
DC reverse protection	Yes
Over current protection	Yes
Over voltage protection	Yes
Anti-islanding protection	Yes
SPD (DC/AC)	Type II (Optional: Type I+II)
Arc-fault circuit interrupter(AFCI)	Optional
AC terminals over temperature detection	Yes
Nighttime SVG	Yes
Fan self-cleaning	Yes
AC auxiliary power supply(APS)	Yes
Power line communication(PLC)	Optional



X1-Micro (2-in-1)

The SolaX X1-Micro series is renowned for its exceptional power output, positioning itself as one of the top-rated models among 2-in-1 microinverters, boasting an impressive capacity of up to 1200 VA. Specifically designed to cater to the latest and previous generation of high-power modules, these microinverters feature two independent MPPTs and robust support for input current and output power. Seamlessly incorporating Wi-Fi wireless technology, the X1-Micro series ensures reliable and consistent communication, offering users a smooth monitoring and control experience. Delivering a cost-effective solution, these microinverters are highly suitable for both residential and commercial solar applications. Furthermore, they are fully compatible with the SolaX Hybrid system and seamlessly integrate with various AC coupled systems available in the market, thus enhancing their versatility and adaptability.

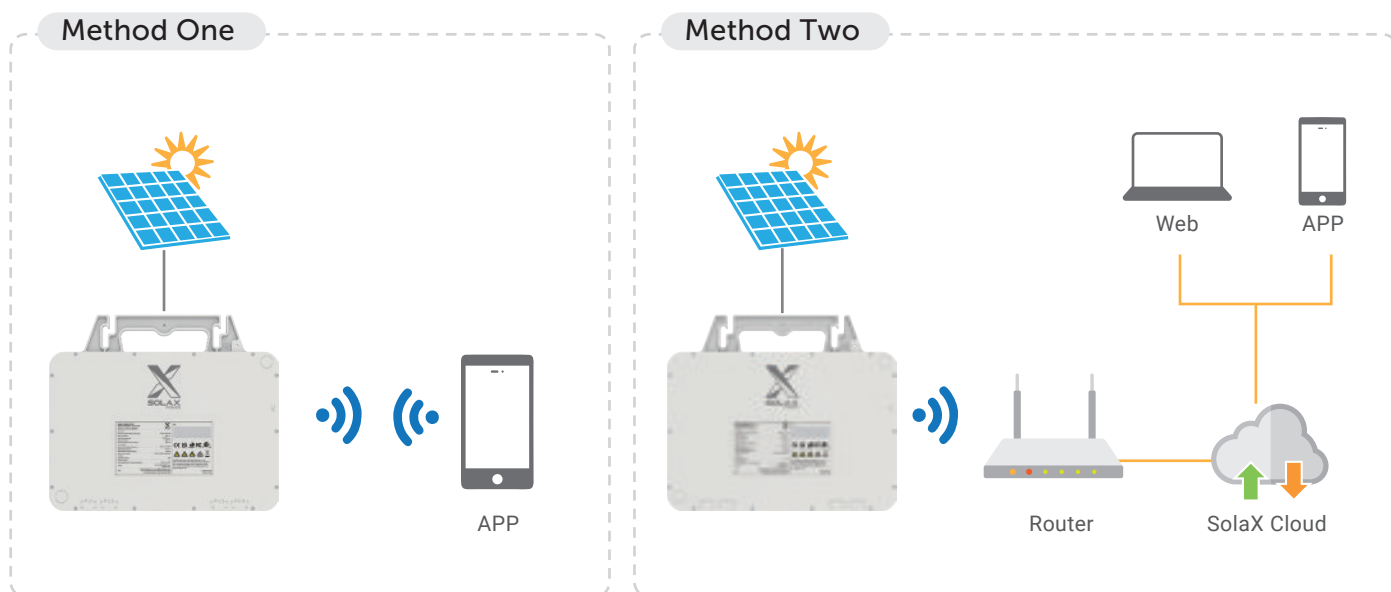


**X1-Micro 800 / X1-Micro 900 /
X1-Micro 1000 / X1-Micro 1200**

Features

- Max output power up to 1200VA with two independent input channels(MPPT)
- Up to 20A DC input current to be compatible with the high power PV module
- Built-in industrial grade Wi-Fi module for high reliability
- Safety protection relay integrated
- Support micro-grid, AC coupling solution with existing storage system
- Support the integration of a single-phase microinverter into a three-phase power grid system
- With Reactive Power Control and Rapid Shutdown Function
- Easy to install and maintain with small size, light weight
- IP67 protection class, more reliable

Communication Diagram



X1-Micro 800

X1-Micro 900

X1-Micro 1000

X1-Micro 1200

DC INPUT

Max. PV input power [W]	320 to 540+	360 to 600+	400 to 670+	400 to 670+
Max. PV input voltage [V]		60		
MPPT voltage range[V]		22-60		
Start-up voltage [V]		22		
Max. PV input current[A]	2 × 15	2 × 16	2 × 20	
Max. input short circuit current Isc [A]	2 × 20	2 × 25	2 × 30	
No. of MPP trackers		2		
Strings per MPP tracker		1		

AC OUTPUT

Rated AC output power [VA]	800	900	1000	1200
Maximum continuous output power [VA]	800	900	1000	1200
Rated AC output current [A]	3.64/3.48/3.34	4.1/3.92/3.75	4.55/4.35/4.17	5.46/5.22/5.00
Rated grid voltage/range[V]		220 or 230 or 240 / 180-264		
Nominal AC frequency/range [Hz]		50/45-55 or 60/55-65		
Maximum units per branch 25A	6/7/7	6/6/6	5/5/5	4/4/5
Power Factor range		>0.99(-0.8~0.8 adjustable)		
THDi (Rated power) [%]		<3%		

Efficiency

Max. efficiency		96.50%		
Nominal MPPT efficiency		99.9%		
Night power consumption [mW]		< 40		

STANDARD

Safety	IEC62109-1/-2, IEC61000-6-1/-2/-3-4, IEC61000-3-2, IEC61000-3-3, IEC 61727, IEC 62116, IEC 61683			
Grid Connection Standard	EN50549-1:2019, VDE-AR-N 4105: 2018, VFR 2019, RD1699:2011, CEI 0-21:2019, ABNT NBR 16149, ABNT NBR 16150, AS/NZS 4777.2:2015			

ENVIRONMENT LIMIT

Ingress protection rating	IP67 (NEMA 6)
Operating ambient temperature range [°C]	-40 ~ 65
Humidity[%]	0 ~ 100(condensing)
Storage temperature[°C]	- 40 ~ 65

GENERAL

Dimension(W x H x D) [mm]	260 × 212 × 40
Weight [kg]	4.1
Cooling concept	Natural convection
Communication	Built-in Wi-Fi
Monitoring	Solax Cloud



X1-Micro (4-in-1)

The SolaX X1-Micro series of microinverters stands out as one of the most powerful options in the 4-in-1 microinverter category, boasting an impressive output power of up to 2200 VA. These cutting-edge products are specifically designed to handle the latest and previous generation of high-power modules with 4 independent MPPTs, allowing for substantial input current and output power capabilities. The integration of a reliable Wi-Fi wireless solution ensures consistent and dependable communication functionality. Offering a cost-effective solution for both residential and commercial solar applications, these microinverters are fully compatible with SolaX Hybrid systems and can also be seamlessly integrated into AC coupled systems on the market.

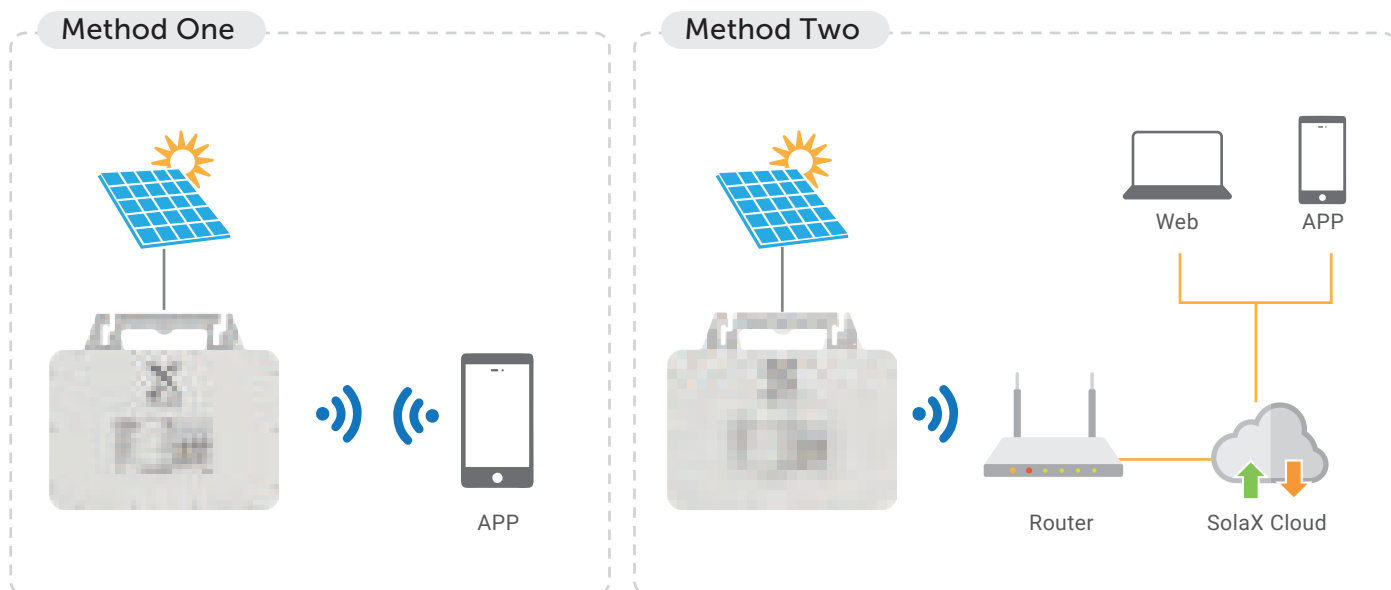


**X1-Micro 1300 / X1-Micro 1500 / X1-Micro 1600 /
X1-Micro 1800 / X1-Micro 2000 / X1-Micro 2200**

Features

- Max output power up to 2200VA with four independent input channels(MPPT)
- Up to 20A DC input current to be compatible with the high power PV module
- Built-in industrial grade Wi-Fi module for high reliability
- Safety protection relay integrated
- Support micro-grid, AC coupling solution with existing storage system
- Support the integration of a single-phase microinverter into a three-phase power grid system
- With Reactive Power Control and Rapid Shutdown Function
- Easy to install and maintain with small size, light weight
- IP67 protection class, more reliable

Communication Diagram



SolaX Microinverter Accessories



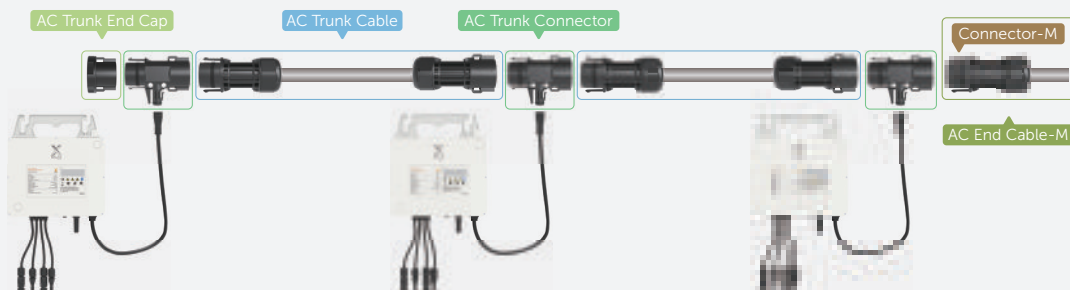
Experience a hassle-free installation with easy, flexible, and reliable cable system

The SolaX microinverter cable system is a revolutionary cabling solution designed specifically for the A1-Micro 1 in 1, X1-Micro 2 in 1 and X1-Micro 4 in 1 series microinverters, which offers a simple and hassle-free installation experience in a PV system with one or multiple microinverters. The plug-and-play design ensures quick and reliable setup, making it easy for users to install. With the flexibility to mix and match connection cables of different sizes and lengths, the intended wiring configuration can be accomplished at a cost-effective price.

Single-Microinverter Solution



Multi-Microinverter Solution



Features



Reliable

- Lower error risks during field installation with Plug-and-play design
- Up to 600 V rated voltage of cable system



Compatible

- Work with Connection Cables of different cable sizes
- Work seamlessly with SolaX Microinverters



Efficient

- Handle current up to 40 A, allowing more microinverters to be connected in trunk cable and lowering system costs
- Improved internal design lowers connection resistance



Simple

- Plug-and-play designs save installation time by 70% and no complex wiring and crimping

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Single-Microinverter Solution

Field Connector



The Field Connector is specifically designed for PV systems with a single microinverter. Used to form the Plug and Play Cable, which facilitates a fast and straightforward electrical connection between the microinverter and the grid, serving as a convenient joining component.

Plug and Play Cable



Plug and Play Cable is specifically designed for PV systems with a single microinverter. It consists of the field connector, AC cable, and plug. The field connector is connected to the microinverter connector, and the plug is connected to the household socket in accordance with local regulations.

Technical Specifications

Connector System Parameter

Pin number	2P + PE
Rated voltage	300 V
Rated current	12 A (Use 1.5 mm ² copper cable)
Applicable cable specification	1.0 / 1.5 / 2.5 mm ²
Applicable cable outer diameter	8.1 mm to 9.6 mm
Cable connection type	screw pressing
Ambient temperature range	-40°C to +85°C
Dimensions (L x W x H)	64 x 28 x 28 mm
Protection degree	IP68
Flame resistance degree	UL94-V0
Compliance	RoHS
Product standard	EN 61984 / UL 6703

Plug and Play Cable Parameter

Consisting components	Field Connector + Cable + AC plug
Cable type	H07RN-F
Conductors cross-sectional area	1.5 mm ²
Cable outer diameter	10.1±0.3 mm
AC Plug standard	EN 50525-2-21
Rated voltage	250 V
Rated current	12 A
Ambient temperature range	-40°C to +85°C
Compliance	RoHS

Multi-Microinverter Solution

Standard	 AC Trunk Cable Consist of an AC Trunk Connector-M, an AC Trunk Connector-F and an AC cable, to connect the AC Trunk Connector	 AC Trunk Connector Consist of an AC Trunk Connector-M, an AC Trunk Connector-F and Microinverter connector, to connect the microinverter's AC output to the AC Trunk, as well as to join together multiple AC Trunk Cables to create the AC Trunk
	 AC Trunk End Cap Used to protect vacant AC Trunk Connector - M	 AC Trunk Port Disconnect Tool A versatile tool that can be used to take apart connectors, tighten nuts, and loosen nuts
Select one	 Connector-M Used to form the AC cable into an AC End Cable or AC Trunk Cable, which completes the connection between the end of the AC Trunk and the distribution box.	 AC End Cable-M Consist of an AC Connector - M and a cable, to connect the microinverter branch to the distribution box
Optional	 Connector-F Used to form the AC cable into an AC End Cable or AC Trunk Cable, which completes the connection between the end of the AC Trunk and the distribution box.	

Technical Specifications

Connector System Parameter	
Pin number	2P + PE
Rated voltage	400 V
Rated current	40 A
Max. supported conductor sizes	6.0 mm ²
Max. supported cable outer diameter	13.5 mm
Ambient temperature range	-40 °C to +85 °C
Protection degree	IP68
Flame resistance degree	UL94-V0
Compliance	RoHS
Product standard	IEC 61984
Cable System Parameter	
Cable type	TC-ER
Rated voltage	600 V
Conductor size	12AWG/10AWG
UV exposure rating	UL1581
Cable flame rating	According to IEC 60332-1-2
Ambient temperature range	-40 °C to +90 °C
Compliance	RoHS
Product standard	E335648(UL)



NEW FROM SOLAX

Hyper EV Charger



X1-HAC-7

X3-HAC-11

X3-HAC-22

Features

- **Green charging**
Integration in new and existing PV systems to charge green energy
- **Automatic phase switching**
Automatic switch between one phase and three phase for maximum utilization of green energy
- **Interoperability**
Support multiple communication protocols to improve interoperability
- **Dynamic load balance**
Realtime monitoring and adjusting to ensure safety
- **Smart APP control**
Smart control and monitoring via Solax APP
- **Easy installation**
3 step to complete installation, easy and fast

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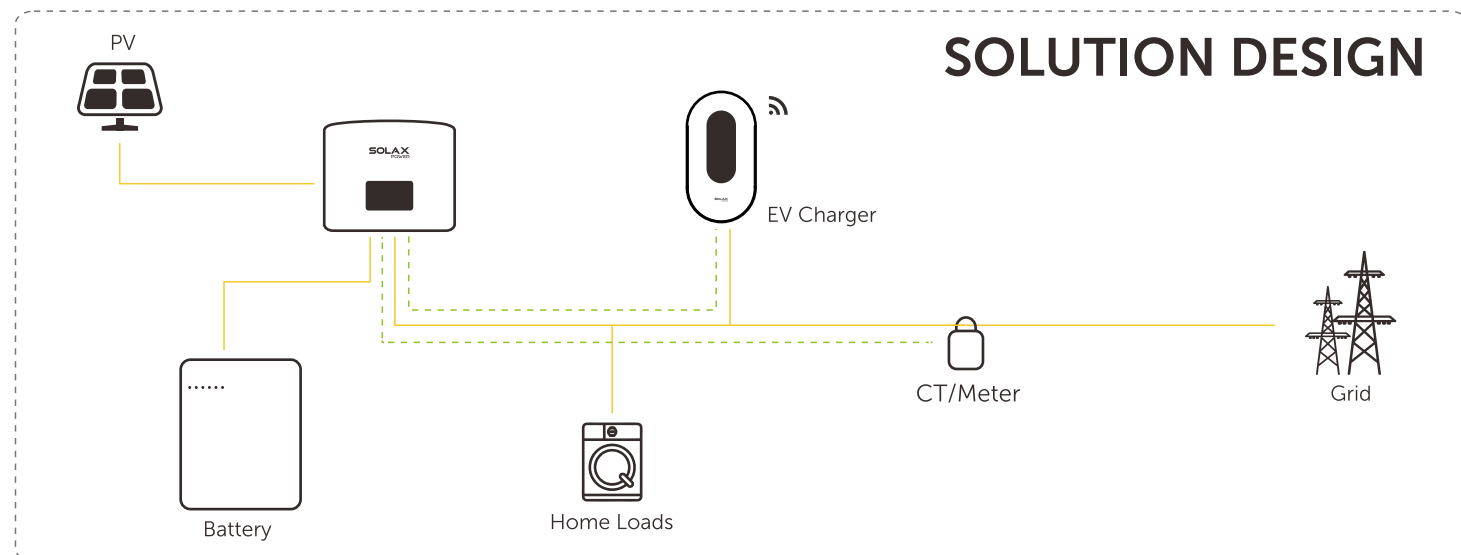
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Hyper EV Charger

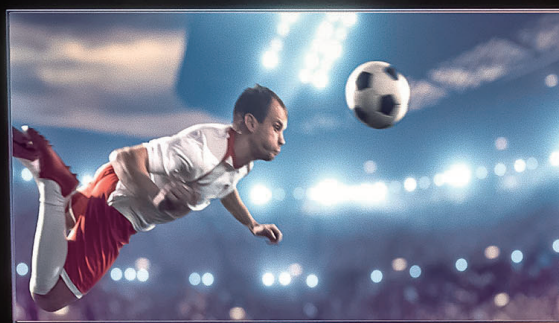
Specification	Model	X1-HAC-7	X3-HAC-11	X3-HAC-22
AC Nominal Input	Voltage [V]	230	400	400
	Frequency [Hz]	50/60	50/60	50/60
	Grid Earth Type	TN, TT, IT	TN, TT	TN, TT
AC Nominal Output	Voltage [V]	230	400	400
	Current [A]	6-32(single phase)	6-16(single phase or three phase)	6-32(single phase or three phase)
	Power [kW]	1.4-7.2	1.4-11	1.4-22
Interface & Communication	Communication interface	WiFi/Ethernet/4G(Optional)RS485		
	Protocol	OCPP 1.6j, Modbus TCP, Modbus RTU, Cloud API		
	Communcate with vehicle	IEC 61851-1, ISO 15118 (Optional)		
	Authentication	Plug&Charge/RFID(ISO-14443-A)/APP		
	MID meter	External(Optional)		
	HMI	RGB LED / APP / LCD(Optional)		
	Remote control	APP&Web		
	Application	Residential/Destination place/Public		
General Data	Housing material	PC		
	Installation method	Wall/Pedestal(optional)		
	Charging outlet	Type P(Charging cable with plug)/Type S(Socket-outlet)		
	Cable length(m)	6.5(Type P)		
	Operating Temperature(°C)	-30~50		
	Storage Temperature(°C)	-40~60		
	Working humidity(%)	5%~95% without condensation		
	Working altitude(m)	2000		
	Degree of Protection	IP65/IK08		
	Cooling method	Nature cooling		
	Application site	Indoor/Outdoor		
	Weight [kg]	3.4 (For socket type)	6.5 (For Plug type)	
	Dimension(WxHxD) [mm]	392*208*139		
Security Protection	Multiple Protection	Over/Under voltage protection,Overload protection,Shortcircuit protection, Current leakage protection,Grounding protection,Surge protection, Overtemperature protection		
	Integral Earth Leakage Protection	Integrated current failure monitoring (30mA AC & 6mA DC)		
	Cable protection	Cable Lock (APP control)		
	Relay protection	Relay weld detection		
	Built-in PEN fault technology	According to BS 7671:2018 requirements*		
	Standard	IEC 61851-1:2017, IEC 62196-2:2016		

*Only for chargers sold in the UK region





NEW FROM SOLAX X1-HYB-LV



X1-HYB-LV

3.0kW/3.7kW/4.0kW
4.6kW/5.0kW/6.0kW

Features

High-efficiency

- 200% PV oversized and up to 110% AC output
- 200% peak EPS apparent power, 10 s
- Built-in shadow tracking

Economic

- Max. DC input current 16A, support for high power solar panel
- Low start up voltage brings longer operation time

Intelligent

- UPS-level switcting time < 10 ms
- CT compatible, loads respond within 0.3 s
- Smart loads management
- Max. 10pcs parallel for on-grid and off-grid operation^①
- Max. charging / discharging current of 120 A
- Generator compatible^②

Safe

- Battery terminal temperature detection
- IP65 protection degree
- Integrated SPD
- AFCI protection (optional)

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X1-HYB-LV

X1-HYB-3.0-LV X1-HYB-3.7-LV X1-HYB-4.0-LV X1-HYB-4.6-LV X1-HYB-5.0-LV X1-HYB-6.0-LV

DC INPUT

Max. PV array power [Wp]	6000	7360	8000	9200	10000	12000
Max. PV input power (PV1+PV2) [Wp]	4500	5500	6000	6900	7500	9000
Max. PV input voltage [V]	550					
Start output voltage [V]	110					
Nominal input voltage [V]	360					
MPPT voltage range [V]	80 ~ 520					
No. of independent MPPT / strings per MPPT	2 / PV1: 1 ; PV2: 1					
Max. input current [A]	16 / 16					
Max. short circuit current [A]	20 / 20					

AC INPUT & OUTPUT

Nominal AC output power [W]	3000	3680	4000	4600	5000	6000
Max. AC output apparent power [VA]	3300	3680	4400	4600 (Germany 4600)	5000	6000
Max. AC output current [A]	15	16	20	20.9 (Germany 20)	22.7	27.3
Max. AC input apparent power [VA]	6000	7360	8000	9200	9200	9200
Max. AC input current [A]	26.1	32	34.8	40	40	40
Nominal voltage [V], frequency [Hz]	220 / 230 / 240, 50 / 60					
Displacement power factor	0.8 leading ~ 0.8 lagging					
THDi (rated power) [%]	< 3					

BATTERY DATA

Battery type	Lithium [®] / Lead-Acid	
Max. charging / discharging current [A]	75	120
Battery voltage range [V]	40 ~ 60	
Nominal battery voltage [V]	48	

EPS OUTPUT (WITH BATTERY)

Nominal output apparent power [VA]	3000	3680	4000	4600	5000	6000
Peak apparent power [VA, s] ^①	6000, 10	7360, 10	8000, 10	9200, 10	10000, 10	12000, 10
Nominal output current [A]	13	16	17.4	20	21.7	26.1
Nominal voltage [V], frequency [Hz]	230, 50 / 60					
Switch time [ms]	< 10					

Efficiency

MPPT Efficiency [%]	> 99.9					
Max. efficiency [%]	97.6					
Euro. efficiency [%]	97.0					

POWER CONSUMPTION

Self consumption (night) [W]	Standby < 40, Shutdown < 10					
------------------------------	-----------------------------	--	--	--	--	--

ENVIRONMENT LIMIT

Degree of protection	IP65		
Operating temperature range [°C]	-25 ~ +60 (derating above +45)		
Relative humidity [%]	0 ~ 100 (condensing)		
Max. operation altitude [m]	< 3000		
Noise emission(typical) [dB]	< 39		< 50
Storage temperature [°C]	-25 ~ +70		

GENERAL

Dimensions (WxHxD) [mm]	397 x 490 x 201	
Net weight [kg]	16.5	17.3
Cooling concept	Natural	Smart cooling
Topology	Transformerless for PV side / HF for battery side	
HMI Interface	LED + LCD	
Communication interfaces	CAN, RS485, CT, Meter, USB, NTC, WiFi, LAN, 4G (Optional), WiFi+Lan (Optional), WiFi+4GM (Optional)	
Warranty [years]	5	

STANDARD

Safety	EN IEC 62109-1 / -2					
EMC	BS EN 50065-1					
Certification	NRS 097-2-1, IEC 61727, IEC 62116, PEA, MEA, BIS					

① To be released in Q3 2023;

② & ③ To be released in Q4 2023;

④ Depend on PV and battery capacity.



NEW FROM SOLAX

X1-Lite-LV



X1-Lite-LV

8.0kW/10.0kW/12.0kW

Features

High-efficiency

- 200% PV oversized and up to 110% AC output
- 200% peak EPS apparent power, 10 s
- Built-in shadow tracking

Economic

- Max. DC input current 16A, support for high power solar panel
- Low start up voltage brings longer operation time

Intelligent

- Single unit UPS-level swtcting time < 4 ms
- CT compatible, loads respond within 0.3 s
- Smart loads management
- Max. 10pcs parallel for on-grid and off-grid operation^①
- Max. charging / discharging current of 250 A
- Generator compatible^②

Safe

- Battery terminal temperature detection
- IP65 protection degree
- Integrated SPD
- AFCI protection (optional)

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X1-Lite-8.0-LV

X1-Lite-10.0-LV

X1-Lite-12.0-LV

DC INPUT

Max. PV Input Power [Wp]	16000	20000	24000
Max. PV Input Voltage [V]		600	
Start Output Voltage [V]		110	
Nominal Input Voltage [V]		360	
Operating Voltage Range [V]		50 ~ 550	
MPPT Voltage Range [V]		50 ~ 550	
No. of MPPT / Strings per MPPT	2 (2 / 2)	2 (2 / 2)	3 (2 / 2 / 2)
Max. Input Current [A]	32 / 32	32 / 32	32 / 32 / 32
Max. Short Circuit Current [A]	40 / 40	40 / 40	40 / 40 / 40

AC INPUT & OUTPUT

Nominal AC Output Power [W]	8000	10000	12000
Max. AC Output Apparent Power [VA]	8800	11000	13200
Max. AC Output Current [A]	40	50	60
Max. AC Input Apparent Power [VA]	12650	14950	18400
Max. AC Input Current [A]	55	65	80
Nominal AC voltage [V], Frequency [Hz]	L / N / PE, 220 / 230 / 240, 50 / 60		
Displacement Power Factor	0.8 leading to 0.8 lagging		
THDi (rated power) [%]	< 3		

BATTERY DATA

Battery Type	Lithium / Lead-Acid		
Battery Voltage Range [V]	40 ~ 60		
Nominal Battery Voltage [V]	48		
Max. Charging / Discharging Current [A]	190	220	250

EPS OUTPUT (WITH BATTERY)

Nominal Output Apparent Power [VA]	8000	10000	12000
Peak Apparent Power [VA, s] ^①	16000, 10	20000, 10	32000, 10
Nominal Output Current [A]	36.4	45.5	54.5
Nominal Voltage [V], Frequency [Hz]	230, 50 / 60		
Switch Time [ms]	< 4		
THDv (linear load) [%]	< 2		

SYSTEM DATA

MPPT Efficiency [%]	> 99.9		
Max. Efficiency [%]	97.6		
Euro. Efficiency [%]	97.0		
Self Consumption(night) [W]	< 10		

ENVIRONMENT LIMIT

Degree of Protection	IP65		
Operating Temperature Range [°C]	-25 ~ +60 (derating above +45)		
Relative Humidity [%]	4 ~ 100 (condensing)		
Max. Operation Altitude [m]	< 3000		
Storage Temperature range [°C]	-25 ~ +70		
Noise Emission(typical) [dB]	< 45		

GENERAL

Dimensions(WxHxD) [mm]	652.5 x 462.5 x 259.8		
Net Weight [kg]	37.4		
Cooling Concept	Smart Cooling		
Topology	Transformerless for PV Side / HF for Battery Side		
HMI / Communication Interfaces	LED + LCD / CAN, RS485, CT, Meter, USB, NTC, WiFi+Lan		
Warranty [years]	5		

STANDARD

Safety	EN IEC 62109-1 / -2		
EMC	BS EN 50065-1		
Certification	NRS 097-2-1, IEC 61727, IEC 62116, PEA, MEA, BIS		

① To be released in Q2 2024;
 ② To be released in Q3 2024;
 ③ Depend on PV and battery capacity.

*V1.0 Information may be subject to modify without notice.

NEW FROM SOLAX X3-NEO-LV



X3-NEO-LV

5.0kW/8.0kW/10.0kW/12.0kW/15.0kW

Features

High-efficiency

- 200% PV oversized and up to 110% AC output
- 200% peak EPS apparent power, 10 s
- Built-in shadow tracking

Economical

- Max. DC input current 18A per channel, supports high power solar panels
- Low start up voltage brings longer operation time

Intelligent

- Single unit UPS-level switching time < 6 ms
- CT compatible, loads respond within 0.3 s
- Max. 10 units parallel for on-grid and off-grid operation^①
- Max. charging / discharging current of 300 A
- Generator compatible^②

Safe

- IP65 protection degree
- Integrated SPD
- AFCI protection(optional)

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DC INPUT

Max. PV Input Power [Wp] ^③	10000	16000	20000	24000	30000
Max. PV Input Voltage [V] ^④			1000		
Start Output Voltage [V]			150		
Nominal Input Voltage [V]			640		
Operating Voltage Range [V] ^⑤			160 ~ 950		
MPPT Voltage Range [V]			160 ~ 950		
No. of MPPT / Strings per MPPT	2 / (2 / 1)			2 / (2 / 2)	
Max. Input Current [A] ^⑥	36 / 18			36 / 36	
Max. Short Circuit Current [A]	50 / 25			50 / 50	

AC INPUT & OUTPUT

Nominal AC Output Power [W]	5000	8000	10000	12000	15000
Max. AC Output Apparent Power [VA]	5500	8800	11000	13200	16500
Max. AC Output Current [A]	8.4	13.4	16.8	20.0	25.0
Max. AC output current [A]@230V/400V	8.0	12.9	16.0	19.2	24.0
Max. AC Input Apparent Power [VA]	10000	16000	20000	24000	30000
Max. AC Input Current [A]	15.3	24.3	30.4	36.5	45.6
Nominal AC voltage [V], Frequency [Hz]	3L / N / PE; 220 / 380, 230 / 400; 50/60				
Displacement Power Factor	0.8 leading ~ 0.8 lagging				
THDi (rated power) [%]	< 3				

BATTERY DATA

Battery Type	Lithium / Lead-Acid				
Battery Voltage Range [V]	40 ~ 60				
Nominal Battery Voltage [V]	48				
Max. Charging / Discharging Current [A]	125	200	235	280	300

EPS OUTPUT (WITH BATTERY)

Nominal output power [W]	5000	8000	10000	12000	15000
Peak Apparent Power [VA , s] ^⑦	10000, 10	16000, 10	20000, 10	24000, 10	30000, 10
Nominal AC Output Current[A]	7.2	11.6	14.5	17.4	21.7
Nominal Voltage [V], Frequency [Hz]	220/380, 230 /400, 50/60				
Switch Time [ms]	< 6				
THDv (linear load) [%]	< 2				

SYSTEM DATA

MPPT Efficiency [%]	> 99.9				
Max. Efficiency [%]	97.6				
Euro. Efficiency [%]	97.0				
Self Consumption [W]	< 15				

ENVIRONMENT LIMIT

Degree of Protection	IP65				
Operating Temperature Range [C]	-25 ~ +60 (derating above +45)				
Relative Humidity [%]	4 ~ 100 (condensing)				
Max. Operation Altitude [m]	< 3000				
Storage Temperature [C]	-25 ~ +70				
Noise Emission (typical) [dB]	≤ 55				

GENERAL

Dimensions(WxHxD) [mm]	706 x 491 x 263				
Net Weight [kg]	43				
Cooling Concept	Smart Cooling				
Topology	Transformerless for PV side / HF for battery side				
HMI / Communication Interfaces	LED+LCD / CAN, RS485, CT, Meter, USB, NTC, WiFi+Lan				
Warranty [years]	5				

STANDARD

Safety	EN IEC 62109-1 / -2 , EN IEC62040				
EMC	EN61000-6 / 1 / 2 / 3 / 4				
Certification	NRS 097-2-1, IEC 61727, IEC 62116, PEA, MEA, BIS				

① ② To be released in Q4 2024; ③ Derating occurs above +45 C., reducing the maximum PV input power;

*V1.0 Information may be subject to modify without notice.

④ The maximum input voltage is the upper limit of the DC voltage. Any higher input DC voltage would probably damage inverter;

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⑤ Input voltage exceeding the operating voltage range may triggers inverter protection;

⑥ When one MPPT has two string inputs, if both strings are connected to the PV, the maximum current per string must not exceed 18A. If only one PV string is connected, the maximum current per string must not exceed 20A.

⑦ Depend on PV and battery capacity.

Disclaimer: The values presented above were obtained in controlled conditions by SolaX's laboratory. Actual performance may differ due to product variations, software updates, operating conditions, and environmental influences.

BY SOLAX
TRIPLE
POWER

NEW FROM SOLAX

T-BAT-SYS-LV D53



T-BAT-SYS-LV D53



Features

Flexibility

- Modular design, expandable to 16 units in parallel, max. 85kWh capacity
- Available in wall-mounted or floor-mounted options

High Performance

- Automotive-grade communication for exceptional hardware performance
- Adaptive cut-off enables continuous operation in parallel, even if any battery fails

Smart Management

- AI-Driven real-time monitoring for minimized system risk
- AI-Driven intelligent algorithms ensure high SOC and accuracy
- AI-Driven battery fault diagnosis for accurate detection

User Friendly

- Continuous remote OTA one-click upgrades

Safe and Durable

- Cobalt-Free LFP battery: safer, long-lasting, efficient, high energy density

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T-BAT-SYS-LV D53

System Paramaters	T-BAT LD53	T-BAT LD106	T-BAT LD159	T-BAT LD212	T-BAT LD265	T-BAT LD318	T-BAT LD371	T-BAT LD424
Number of Module Serie	1	2	3	4	5	6	7	8
Nominal Capacity [kWh]	5.3	10.6	15.9	21.2	26.6	31.9	37.2	42.5
Usable Capacity (90% DOD) ^① [kWh]	4.7	9.5	14.3	19.1	23.9	28.7	33.5	38.3
Max. Output Current [A] ^②	100	120						
Peak Discharge Current [A, s]	200, 10							

System Paramaters	T-BAT LD477	T-BAT LD530	T-BAT LD583	T-BAT LD636	T-BAT LD689	T-BAT LD742	T-BAT LD795	T-BAT LD848
Number of Module Serie	9	10	11	12	13	14	15	16
Nominal Capacity [kWh]	47.9	53.2	58.5	63.8	69.2	74.5	79.8	85.1
Usable Capacity (90% DOD) ^① [kWh]	43.1	47.9	52.7	57.5	62.3	67.0	71.8	76.6
Max. Output Current [A] ^②	120							
Peak Discharge Current [A, s]	200, 10							

General Information	
Weight [kg]	48.5
Dimension (L x W x H) [mm]	645 x 150 x 430
Nominal Voltage [V]	51.2
Operating Voltage Range [V]	45 to 58
Battery Type	Lithium Iron Phosphate
Communication Port	CAN / RS485
Operation Temperature [°C]	0 to 53 (charge) ; -20 to 53 (discharge)
Storage Temperature [°C]	30 to 50 (6 months) ; -20 to 30 (12 months)
Ingress Protection	IP65
Colling concept	Natural
Relative humidity [%]	5 to 95 (Non-condensing)
Altitude [m]	< 3000
Warranty ^④ [years]	10
Cycle Life ^③ [90% DOD]	> 6000
Certification	IEC62619, IEC62040, CE, UN38.3

① Test conditions: 90% DOD, 0.2C charge & discharge @+25 °C

② Current is affected by the number of batteries connected in parallel as well as temperature and SOC

③ 25°C ± 2°C, 0.5C / 0.5C, 70 % EOL > 6000

④ The warranty is due whichever reached first of warranty period or energy throughput

BY SOLAX
TRIPLE
POWER

NEW FROM SOLAX T-BAT-SYS-LV D150



T-BAT-SYS-LV D150



Features

- Safe LiFePO4 battery
- Peak discharge current: 310A, 10s
- Long cycle life > 6000 times
- Easy and Fast installation
- Remote monitoring and upgrade
- Easy to scale up, support Max. 16 PCS LD150 parallel

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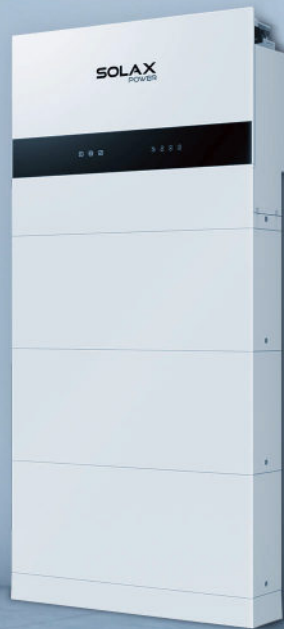
T-BAT-SYS-LV D150

Type

Battery Type	Lithium Iron Phosphate
Battery Component	1P5S*3
Nominal Voltage [V]	48
Operating Voltage Range [V]	42 to 54
Rated Capacity [Ah]	314
Nominal Energy [kWh]	15.0
Usable Energy (90% DOD) ^① [kWh]	13.5
Max. Charge / Discharge Current [A]	155
Peak Discharge Current [A,s]	310A, 10s
Cycle Life ^②	> 6000 Cycles
Communication Interfaces	CAN / RS485
Operation Temperature [°C]	0 to 53 (charge) / -20 to 53 (discharge)
Storage Temperature [°C]	-20 to 50 (3 months) / 10 to 40 (1 year)
Max. Parallel Number [PCS]	16
Dimension (LxWxH) [mm]	900*540*260
Weight [kg]	130
Installation Type	Floor
Degree of protection	IP65
Cooling concept	Natural
Warranty [years]	5
Max. operation altitude [m]	< 3000
Certification	UN38.3 / IEC62619 / CE

① ② Test conditions: 90% DOD, 0.2C charging & discharge @+25 °C.

SOLAX X1-IES



X1-IES

3kW/3.7kW/4.6kW/5kW/6kW/8kW



INTRODUCTION

This is an integrated residential ESS which comes with a 3–8kW hybrid single-phase inverter and extensible battery modules, plug and play, capacity range from 5 to 20 kWh. It has excellent performance in economy, safety and robustness. In addition, intelligent functions like VPP, micro-grid, smart schedule and smart scene are all ready. It would be the best choice for householders.

SolaX IES

◀ Features

Economic

- All in one design, plug and play, expandable and installation easily
- Maximum 200% oversize and 200% PV input power
- Maximum 20A DC single string input current, support high power solar panel
- Low start output voltage makes inverter longer working time
- Built-in shadow tracking function
- Maximum Charge / Discharge Current up to 50A

Safe

- IP66 protection level
- AC&DC SPD type II, always guarding the inverter
- AFCI optional

Robust

- Robust back-up ability, switchover time <10ms(UPS level), up to 200% EPS output for 10s
- Battery heating technology, -30°C extreme environment operation

Intelligent

- AI ready, forecasting solar generation and home consumption, smart energy management strategy
- VPP ready, SolaX cloud support resource aggregator(IEEE 2030.5, OpenADR)
- Micro-grid ready, supporting real-time power balance between PCS and Hybrid in grid-connected and off-grid scenarios.
- Support smart scene function, intelligent loads management (e.g., Heat pump, EV charger)
- Support 7×24h scheduling mode
- Support Wireless meter solution



SYSTEM OVERVIEW

System schematic



Rated output power [kW]	2 / 3 / 3.7 / 4.6 / 5 / 6 / 8			
Number of batteries	1	2	3	4
Nominal capacity [kWh] ^①	5.1	10.2	15.3	20.4
Usable energy [kWh] ^②	4.6	9.2	13.8	18.4
Max. charge/discharge power [kW] ^③	5.1	8	8	8
Degree of protection	IP66			
Operating temperature range [°C]	-30 to 53			
Allowable relative humidity range [%]	5-95 (No condensation)			
Max. operating altitude [m]	3000			
Net weight [kg] ^④	87.2	134.2	181.2	134.2 / 99.2
Dimension (W x H x D) [mm]	730 x 908 x 210	730 x 1226 x 210	730 x 1544 x 210	730 x 1226 x 210 / 730 x 809 x 150
Display	LCD			
Cooling concept	Natural cooling			
Topology	Non-isolated			
Communication	RS485, Pocket-X, USB, CAN, DO, DI			

① Test conditions: 25°C, 100% depth of discharge (DoD), 0.2C charge & discharge.

② System usable energy may vary with inverter different setting.

③ The max.charge/discharge power must not exceed the rated output power (the table takes the maximum power inverter as an example).

④ Different inverter models have different weights. The heaviest one is taken as an example.

SPECIFICATIONS

	X1-IES-2.5K	X1-IES-3K	X1-IES-3.7K	X1-IES-4.6K	X1-IES-5K	X1-IES-6K	X1-IES-8K
INPUT PV							
Max. recommended PV array power [Wp]	5000	6000	7400	9200	10000	12000	16000
Max. DC voltage [V]	600						
Nominal DC operating voltage [V]	360						
Max. input current (input PV1 / input PV2 / input PV3) [A]	PV1: 20 / PV2: 20		PV1: 20 / PV2: 20 / PV3: 20				
Max. short circuit current (input PV1 / input PV2 / input PV3) [A]	PV1: 30 / PV2: 30		PV1: 30 / PV2: 30 / PV3: 30				
MPPT voltage range ^① [V]	40 to 560						
Start output voltage [V]	50						
No. of MPP trackers / Strings per MPP tracker	2 / (1 / 1)	2 / (1 / 1)	2 / (1 / 1)	3 / (1 / 1 / 1)	3 / (1 / 1 / 1)	3 / (1 / 1 / 1)	3 / (1 / 1 / 1)
INPUT AC							
Nominal AC power [VA]	6300	6300	7360	9200	9200	9200	9200
Max. AC current [A]	27.4	27.4	32	40.0	40.0	40.0	40.0
Rated grid frequency [Hz]	50 / 60						
Power factor	~1 (Adjustable from 0.8 leading to 0.8 lagging)						
OUTPUT AC (On-Grid)							
Nominal AC power [VA]	2500	3000	3680	4600	5000 (4600 for VDE4105, 4999 for AS4777)	6000	8000
Max. apparent AC power [VA]	2500	3300	3680	4600	5000 (4600 for VDE4105, 4999 for AS4777, 5000 for C10/11)	6600	8000
Rated grid voltage(AC voltage range) [V]	Single phase, 220 / 230 / 240						
Rated grid frequency [Hz]	50 / 60						
Rated AC output current [A] (at 230V, 50Hz)	10.9	13.1	16	20	21.8	26.1	34.8
Max. AC current [A] (at 230V, 50Hz)	10.9	14.4	16	20	21.8	28.7	34.8
Displacement power factor	~1 (Adjustable from 0.8 leading to 0.8 lagging)						
Total harmonic distortion (THDi, rated power) [%]	< 3						

EPS OUTPUT (With battery)							
EPS peak power [VA]	≤1.1Pn continuous operation; 1.1Pn-2Pn 10s; >2Pn report error immediately						
EPS rated power [VA]	2500	3000	3680	4600	5000	6000	8000
EPS rated voltage [V]; frequency [Hz]	Single phase, 220 / 230 / 240; 50/60						
EPS rated current [A]	10.9	13.1	16	20	21.8	26.1	34.8
Switchover time [ms]	< 10						
Total harmonic distortion (THDv, linear Load) [%]	< 3						
BATTERY							
Battery voltage range [V]	80 ~ 480						
Communication interfaces	CAN / RS485						
BMS module	TBMS-MCS0800E						
Battery module	TP-HS50E						
Composition	TBMS-MCS0800E + TP-HS50E * n + Base Dimensions + Series Box (Required for two columns)						
Battery type	Li-ion (LFP)						
Nominal capacity [kWh] / Nominal capacity [Ah]②	5.1 / 50						
Usable energy [kWh]③	4.6						
Standard power [kW]	3						
Max power [kW]	5.1						
Max. charge / discharge current [A]④	50						
Cycle life [Cycles]	> 6000						
Warranty [Years]	10						
Safety	CE, RCM, TUV (IEC62619), RoHS, REACH						
TBMS-MCS0800E dimensions(W × H × D) [mm] / Weight [kg]	730 × 165 × 150 / 9.3						
TP-HS50E dimensions(W × H × D) [mm] / Weight [kg]	730 × 318 × 150 / 47						
Base dimensions(W × H × D) [mm] / Weight [kg]	730 × 75 × 150 / 3.9						
Series box dimensions(W × H × D) [mm] / Weight [kg]	167 × 91.5 × 121 / 1.3						
EFFICIENCY							
Max. efficiency [%] / Euro-efficiency [%]	97.6 / 97.0						
Rated battery charge [%] / Discharge efficiency [%]	98.5 / 97						
GENERAL DATA (INVERTER)							
Dimensions (W × H × D) [mm]							
Weight [kg]	26.2			26.4			27
Operating temperature range [°C]							
Relative humidity [%]							
Storage temperature [°C]							
Noise emission (typical) [dB(A)]							
Internal consumption (night) [W]							
PROTECTION							
Anti-Islanding protection	Yes						
DC reverse polarity protection	Yes						
Insulation monitoring	Yes						
Residual current monitoring	Yes						
AC overcurrent protection	Yes						
AC short-circuit protection	Yes						
AC overvoltage protection	Yes						
Over-heat protection	Yes						
AFCI	OPT						
Surge protection	Type II, DC and AC						
STANDARD							
Safety	IEC62109-1 / IEC62109-2						
EMC	EN 61000-6-1 / EN 61000-6-2 / EN 61000-6-3						
Certification	VDE 0126-1-1 A1:2012 / VDE-AR-N 4105 /G98/G99/ AS4777 / EN50549 / CEI 0-21						

① Any DC input voltage beyond the MPPT voltage range may result in inverter improper operating.

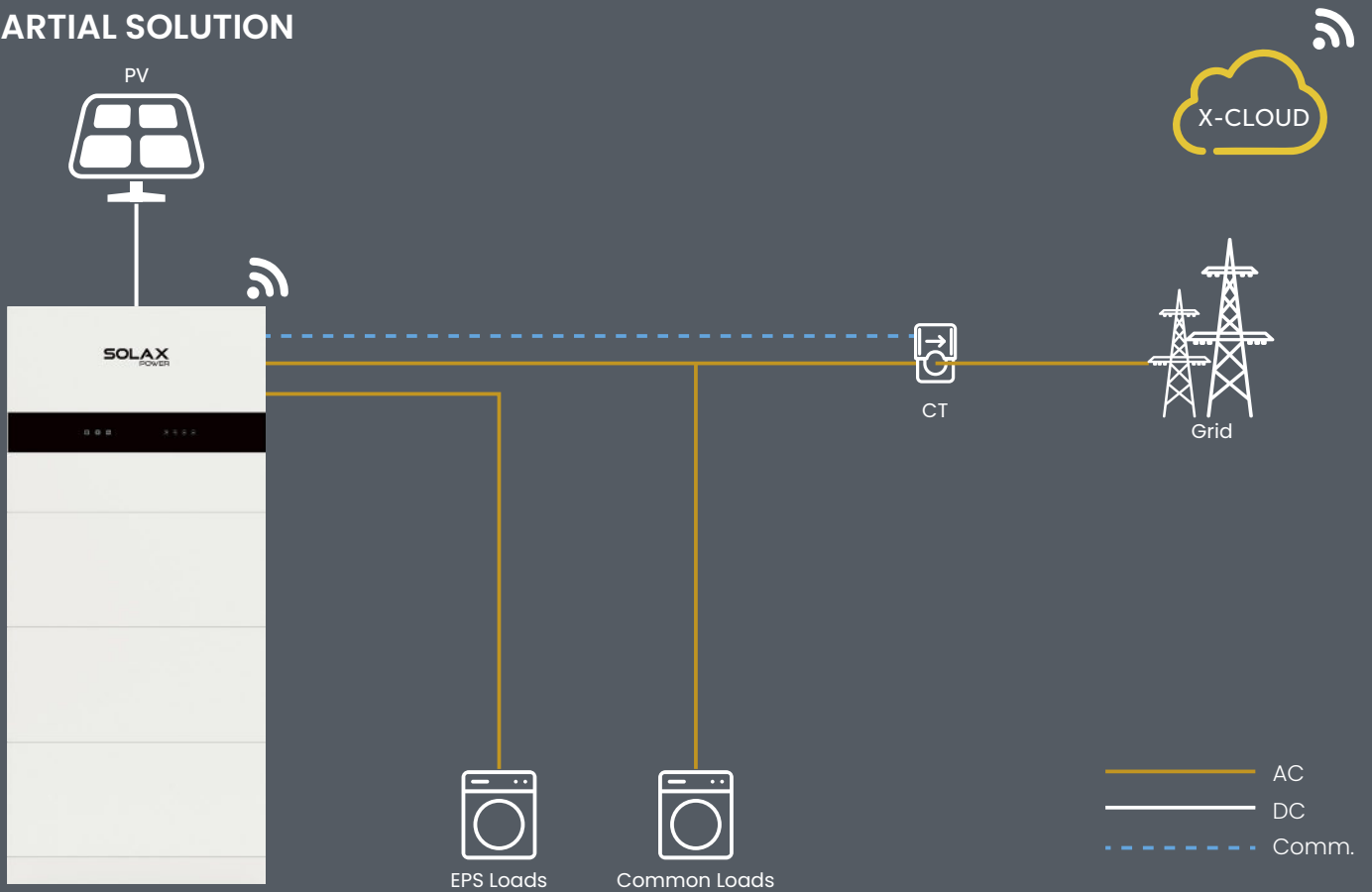
② Test conditions: 25°C, 100% depth of discharge (DoD), 0.2C charge & discharge.

③ System usable energy may vary with inverter different setting.

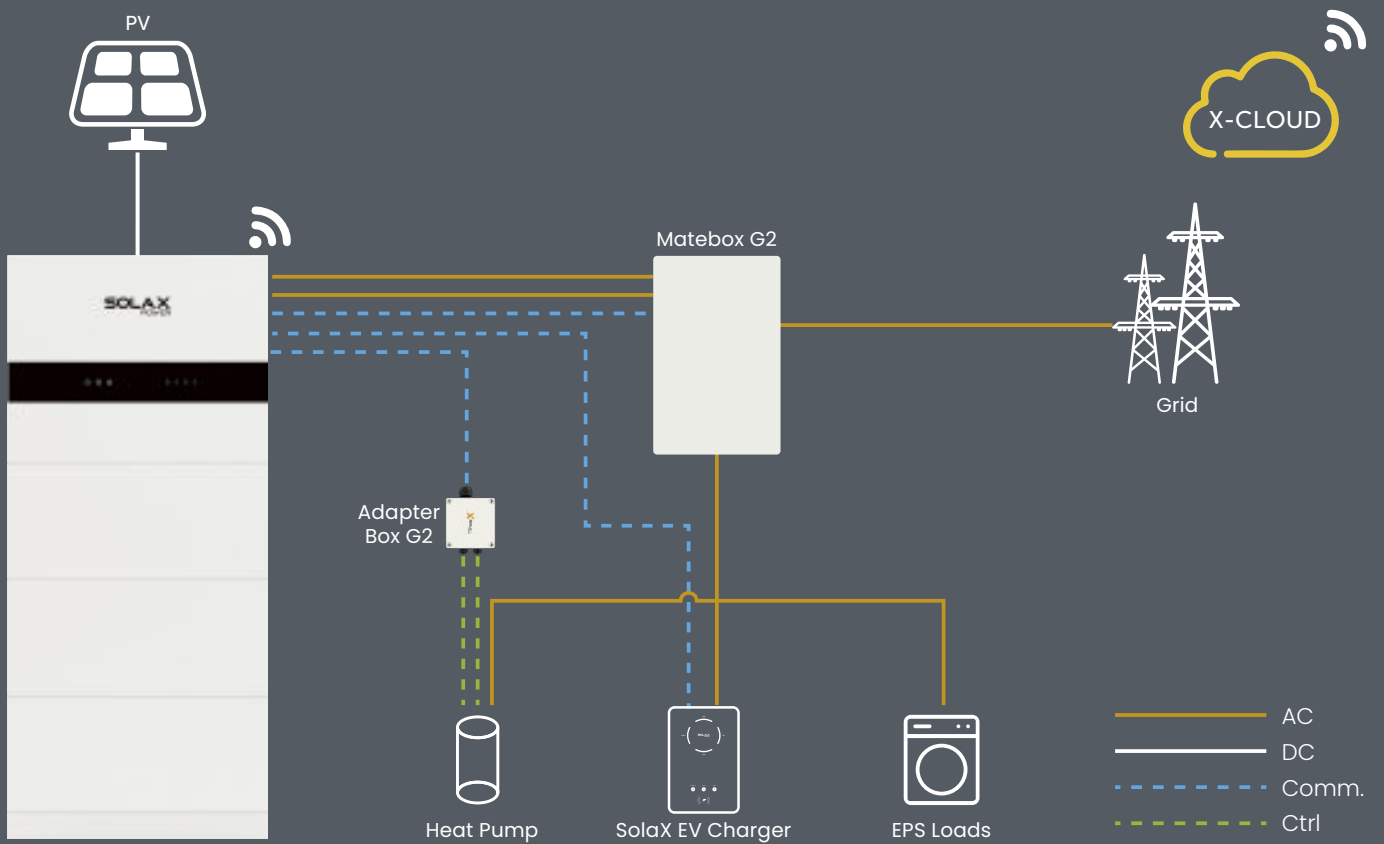
④ Discharge: In case of battery cell's temperature range of -20°C~10°Cand 45°C~53 °C,the discharge current will be reduced; Charge: In case of battery cell's temperature range of 0°C~25°C and 45°C~53°C, the charge current will be reduced. Product charge or discharge power depends on the actual temperature of battery pack.

TACTICAL SCENARIO

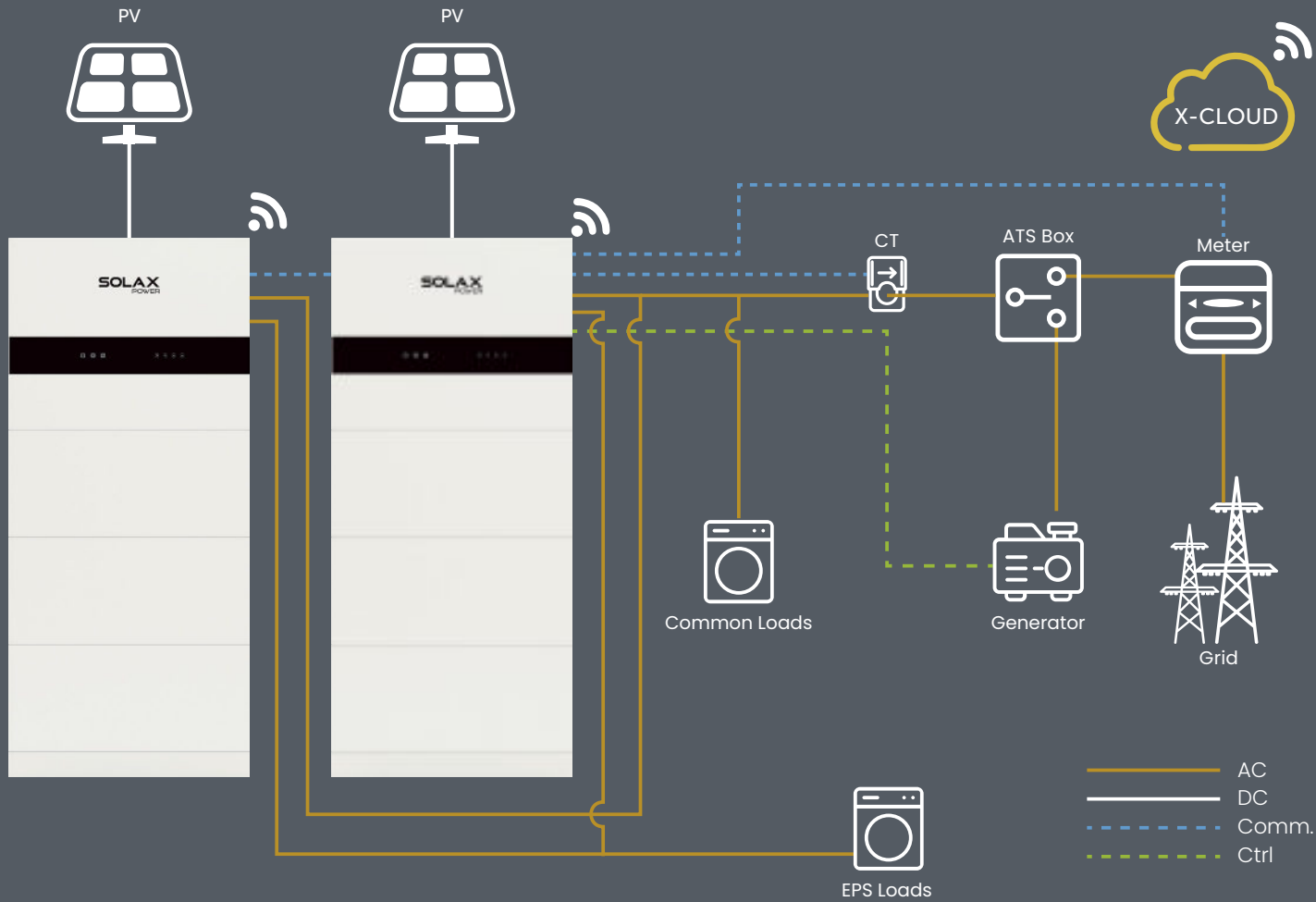
PARTIAL SOLUTION



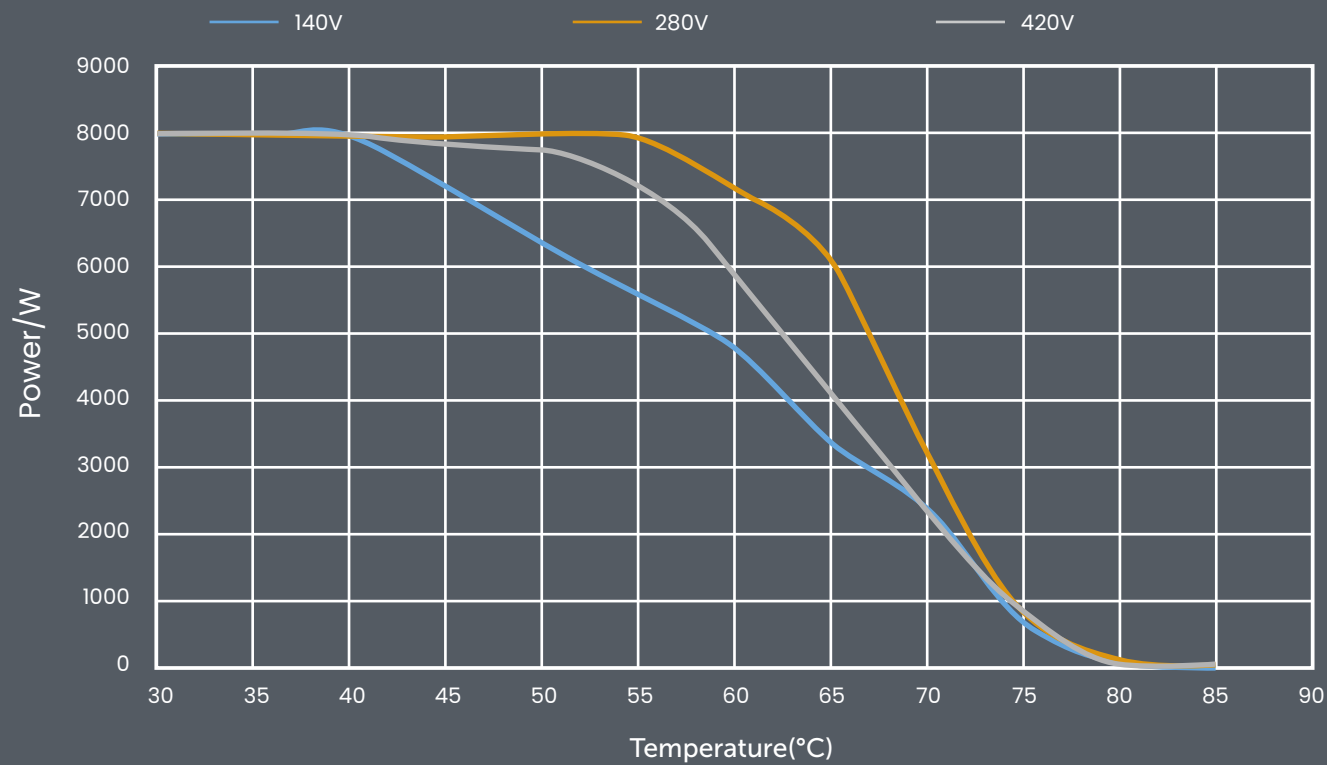
WHOLE HOME BACKUP SOLUTION



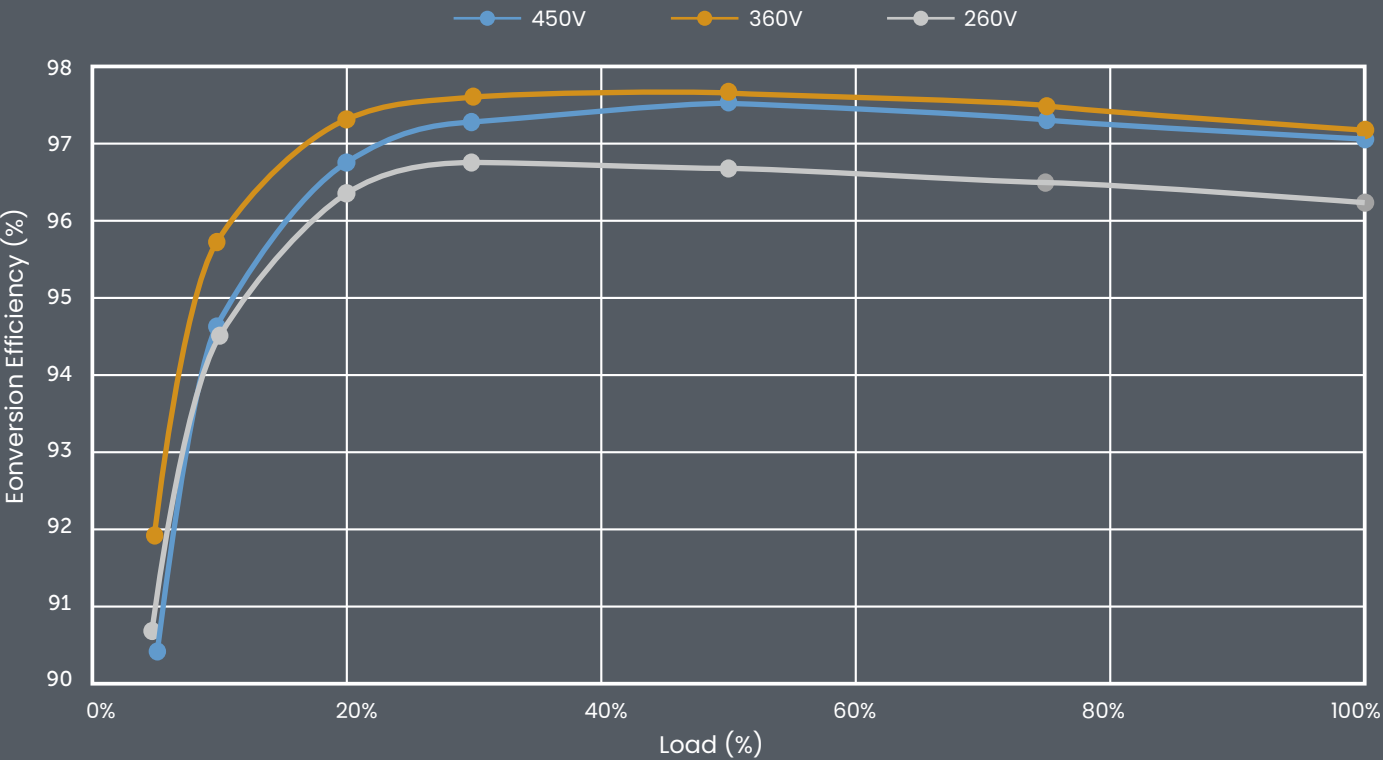
PARALLEL & GENERATOR INTEGRATED SOLUTION



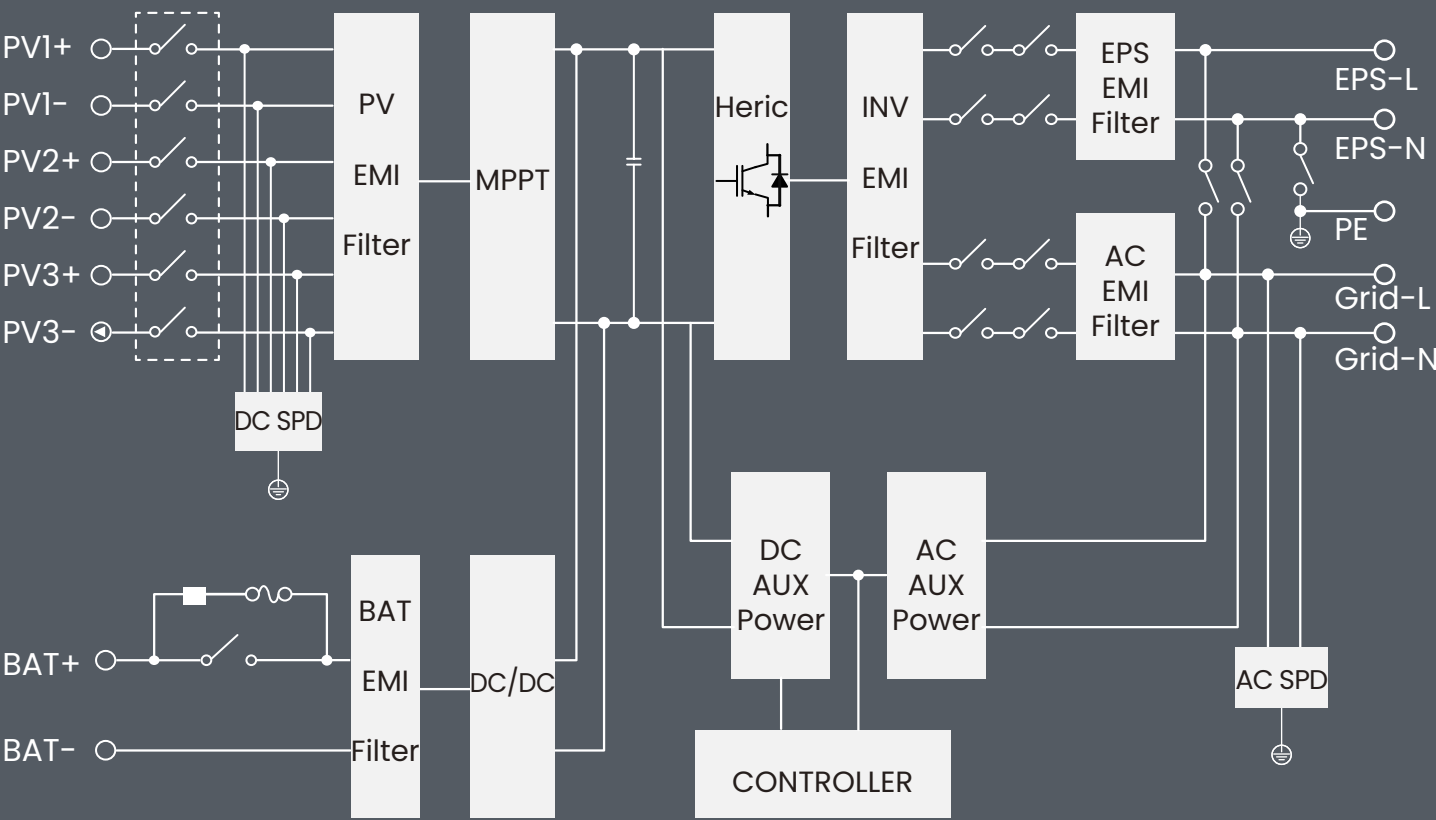
Derating Curve (8kw)



EFFICIENCY CURVE



CIRCUIT DIAGRAM





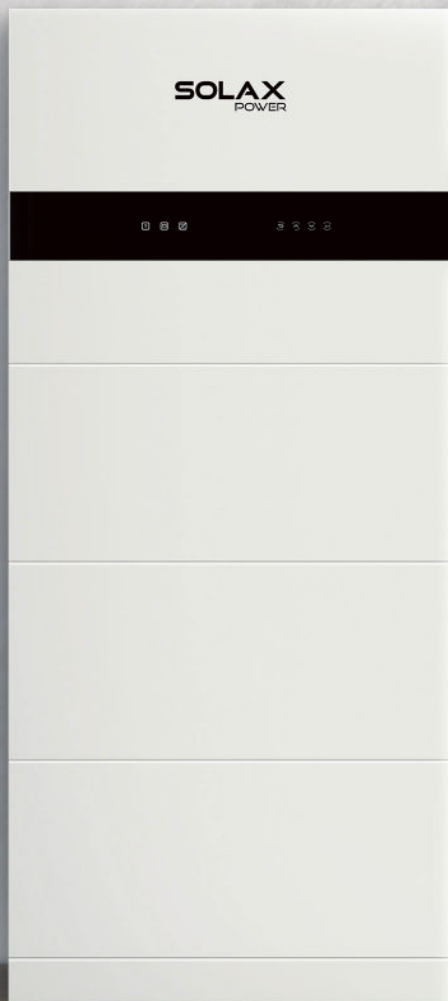
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SOLAX X3-IES



X3-IES

5kW/6kW/8kW/10kW/12kW/15kW





INTRODUCTION

This is an integrated residential ESS which comes with a 5-15kW hybrid Three-phase inverter and extensible battery modules, plug and play, capacity range from 10 to 30 kWh. It has excellent performance in economy, safety and robustness. In addition, intelligent functions like VPP, micro-grid, smart schedule and smart scene are all ready. It would be the best choice for householders.

SolaX IES

◀ Features

Economic

- All in one design, plug and play, expandable and installation easily
- Maximum 200% oversize and 200% PV input power
- Maximum 20A DC single string input current, support high power solar panel
- Low start output voltage makes inverter longer working time
- Built-in shadow tracking function

Safe

- IP66 protection level
- AC&DC SPD type II, always guarding the inverter
- AFCI optional

Robust

- Robust back-up ability, switchover time <10ms(UPS level) , up to 200% EPS output for 10s, support half-wave loads
- Battery heating technology, -30°C extreme environment operation

Intelligent

- AI ready, forecasting solar generation and home consumption, smart energy management strategy
- VPP ready, SolaX cloud support resource aggregator(IEEE 2030.5, OpenADR)
- Micro-grid ready, supporting a variety of scenarios, both on-grid and off-grid, balancing power between PCS and Hybrid in real time.
- Support smart scene function, intelligent loads management(e.g., Heat pump, EV charger)
- Support 7×24h scheduling mode
- Support Wireless meter solution

EPS OUTPUT(With Battery)

EPS peak power [VA]	≤1.1Pn continuous operation; 1.1Pn-2Pn 10s; >2Pn report error immediately					
EPS rated power [VA]	4000	5000	6000	8000	10000	15000
EPS rated voltage [V], Frequency [Hz]	3P4W, 380 / 400, 50/60					
EPS rated current [A]	5.8	7.3	8.7	11.6	14.5	21.8
Switchover time [ms]	< 10					
Total harmonic distortion (THDv, linear Load) [%]	< 3					
Max. half wave loads [kW]	2					

BATTERY

Battery voltage range [V]	160 ~ 800					
Communication interfaces	CAN / RS485					
BMS module	TBMS-MCS0800E					
Battery module	TP-HS50E					
Composition	TBMS-MCS0800E + TP-HS50E * n + Base Dimensions + Series Box (Required for two columns)					
Battery type	Li-ion (LFP)					
Nominal capacity [kWh] / Nominal capacity [Ah] ^②	5.1 / 50					
Usable energy [kWh] ^③	4.6					
Standard power [kW]	3					
Max power [kW]	5.1					
Max. charge / discharge current [A] ^④	50					
Cycle life [Cycles]	> 6000					
Warranty [Years]	10					
Safety	CE, RCM, TUV (IEC62619), RoHS, REACH					
TBMS-MCS0800E dimensions(W x H x D) [mm] / Weight [kg]	730 x 165 x 150 / 9.3					
TP-HS50E dimensions(W x H x D) [mm] / Weight [kg]	730 x 318 x 150 / 47					
Base dimensions(W x H x D) [mm] / Weight [kg]	730 x 75 x 150 / 3.9					
Series box dimensions(W x H x D) [mm] / Weight [kg]	167 x 91.5 x 121 / 1.3					

EFFICIENCY

Max. efficiency [%] / Euro-efficiency [%]	98 / 97.7					
Rated battery charge [%] / Discharge efficiency [%]	98.5 / 97					

GENERAL DATA (Inverter)

Dimensions (W x H x D) [mm]	717 x 405 x 209,5		
Weight [kg]	35	37	
Operating temperature range [°C]	– 35 to 60 (derating at +45)		
Relative humidity [%]	0 to 100 (condensing)		
Storage temperature [°C]	– 40 to 65		
Noise emission (typical) [dB(A)]	< 33		
Internal consumption (night) [W]	< 40 for hot standby, < 5 for cold standby		
Idle mode	Yes		

PROTECTION

Anti-Islanding protection	Yes					
DC reverse polarity protection	Yes					
Insulation monitoring	Yes					
Residual current monitoring	Yes					
AC overcurrent protection	Yes					
AC short-circuit protection	Yes					
AC overvoltage protection	Yes					
Over-heat protection	Yes					
AFCI	OPT					
Surge protection	Type II, DC and AC					

STANDARD

Safety	IEC62109-1 / IEC62109-2					
EMC	EN 61000-6-1 / EN 61000-6-2 / EN 61000-6-3					
Certification	VDE 0126-1-1 A1:2012 / VDE-AR-N 4105 / G98/G99/ AS4777 / EN50549/ CEI 0-21					

① Any DC input voltage beyond the MPPT voltage range may result in inverter improper operating.

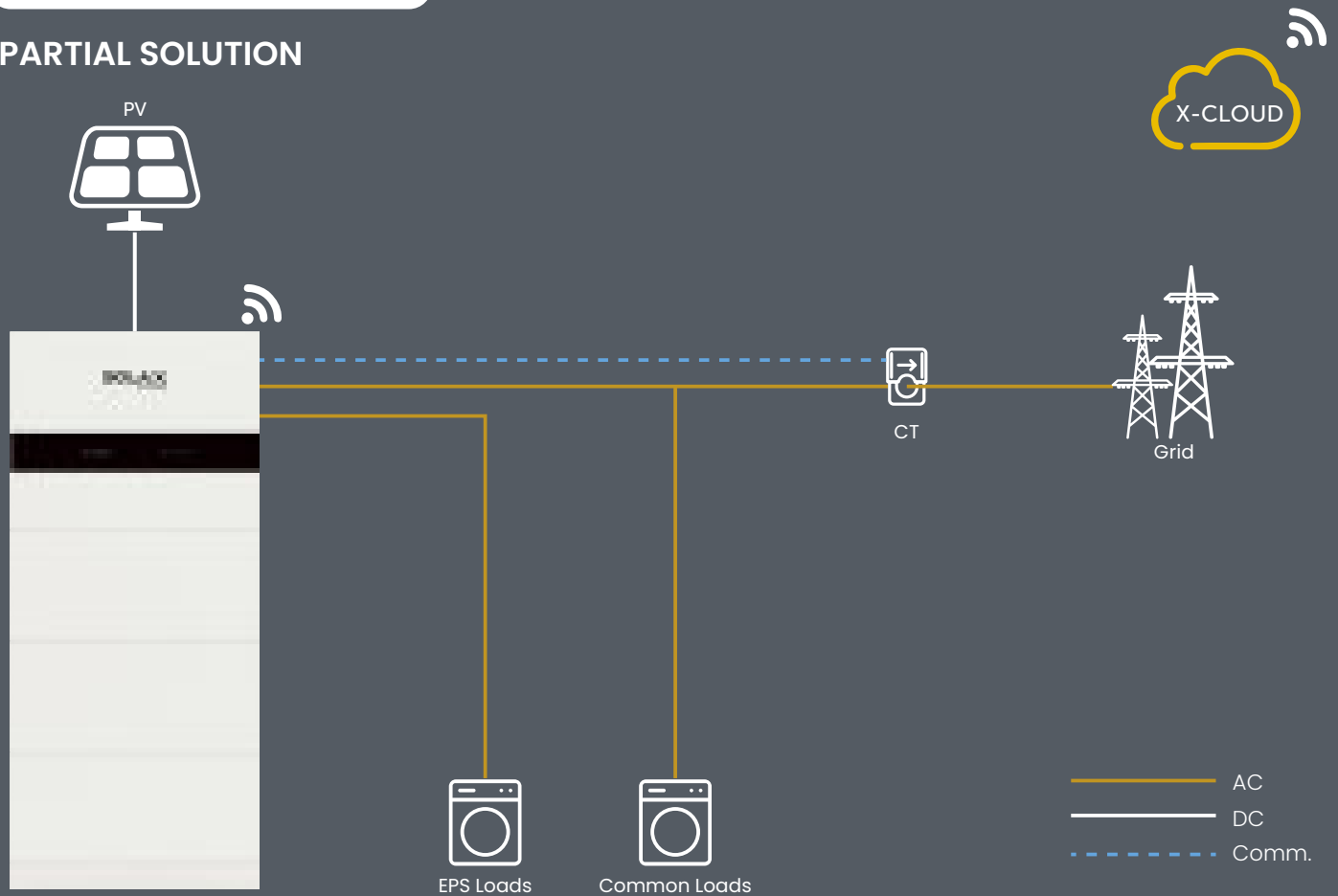
② Test conditions: 25°C, 100% depth of discharge (DoD), 0.2C charge & discharge.

③ System usable energy may vary with inverter different setting.

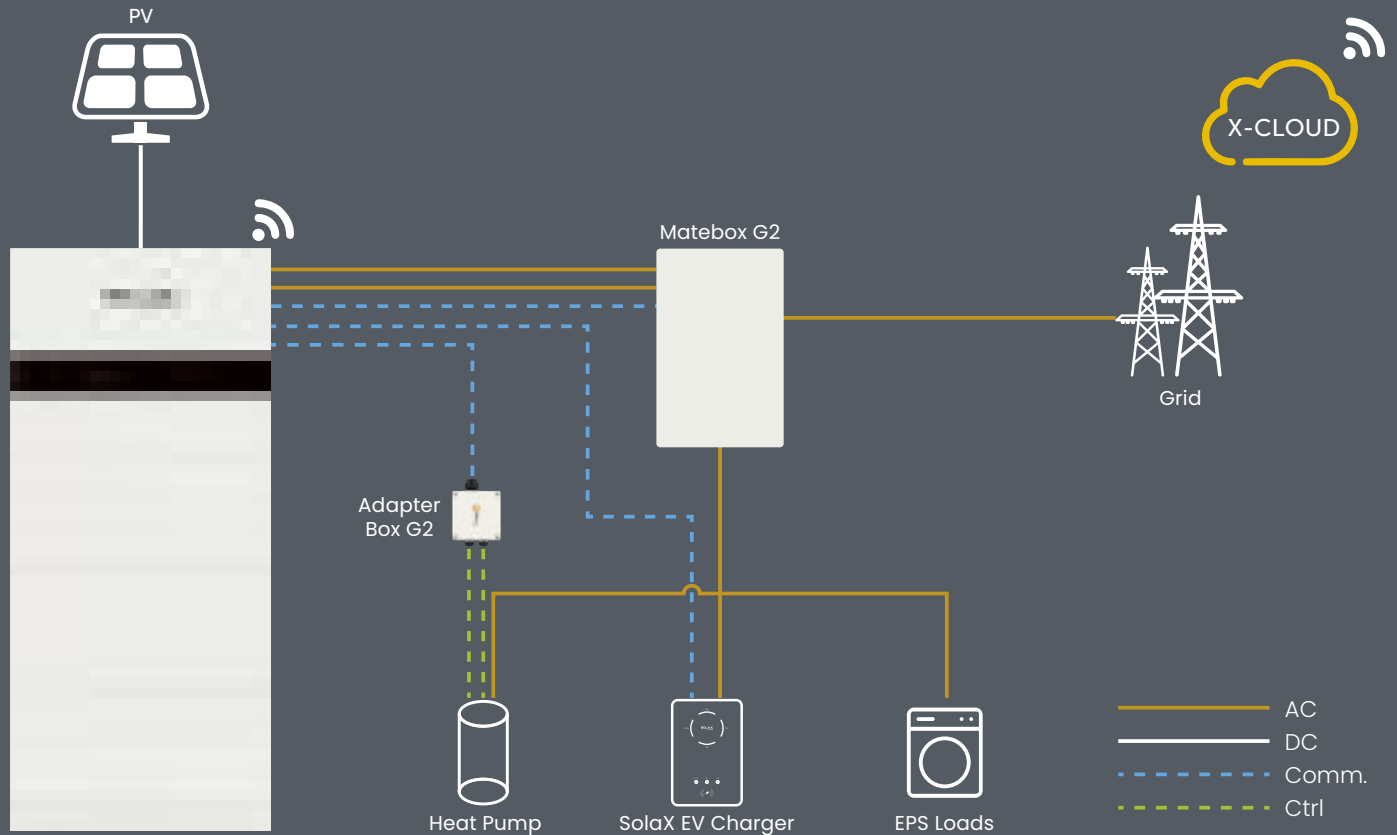
④ Discharge: In case of battery cell's temperature range of -20°C~10°C and 45°C~53 °C, the discharge current will be reduced; Charge: In case of battery cell's temperature range of 0°C~25°C and 45°C~53°C, the charge current will be reduced. Product charge or discharge power depends on the actual temperature of battery pack.

TACTICAL SCENARIO

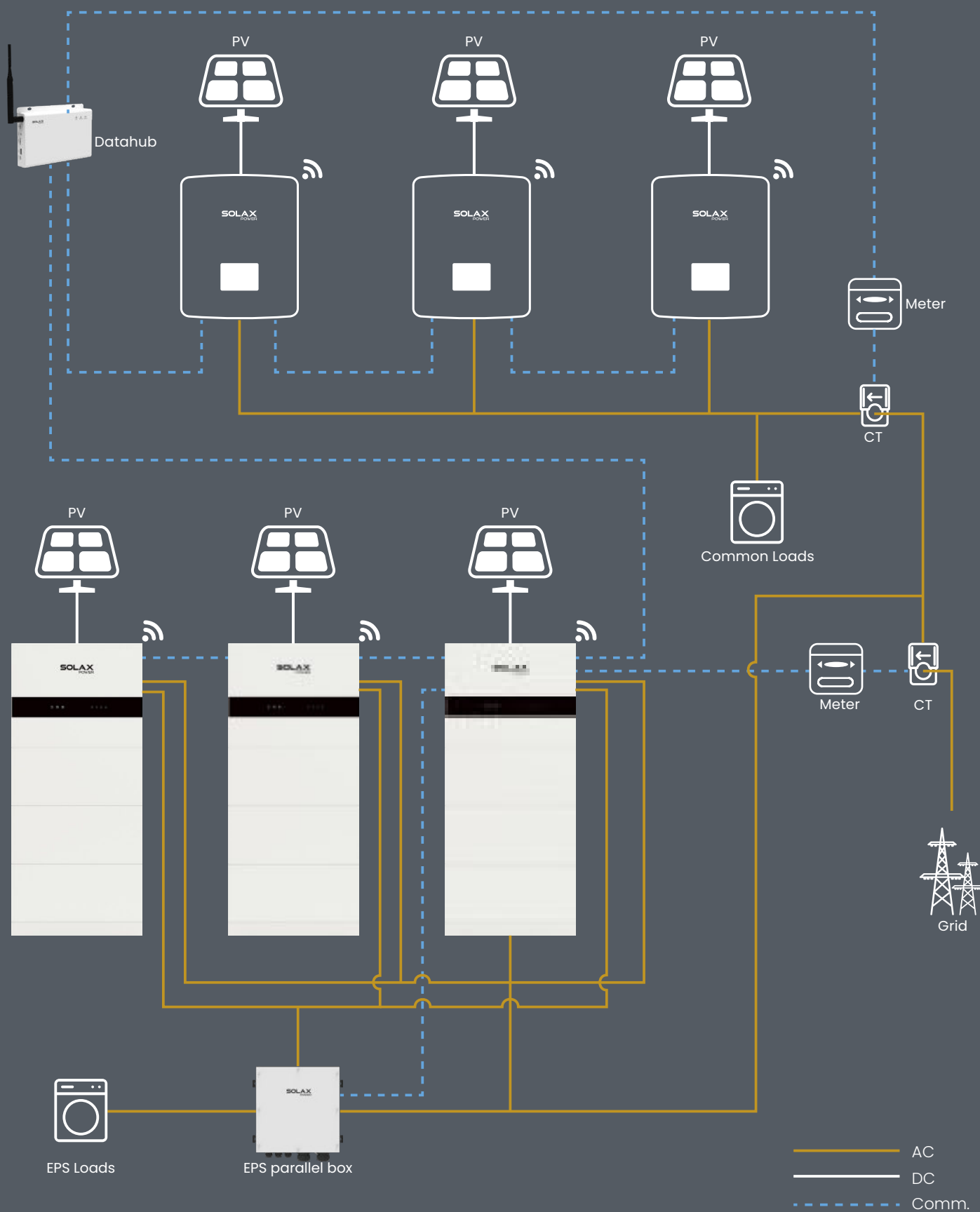
PARTIAL SOLUTION



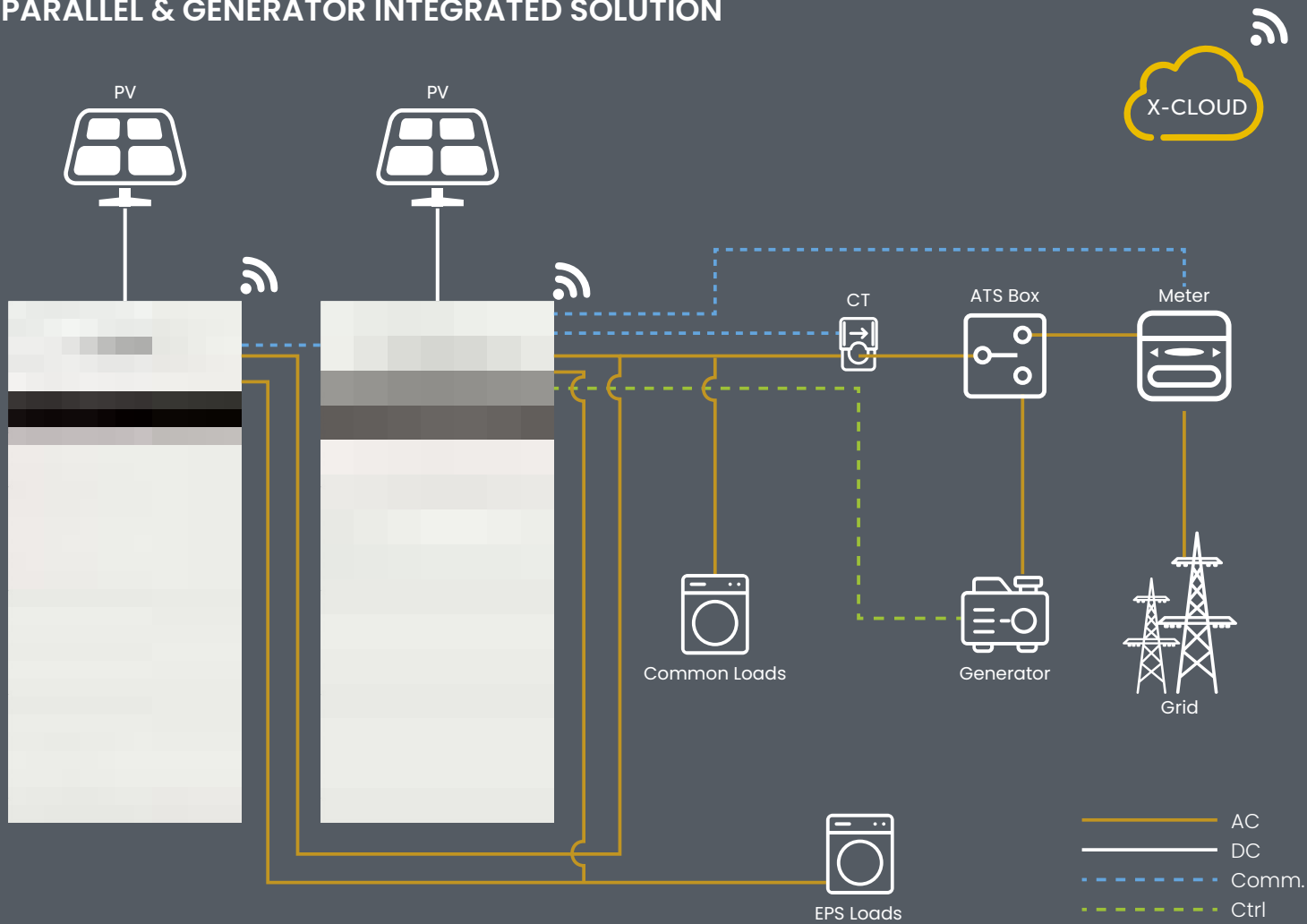
WHOLE HOME BACKUP SOLUTION



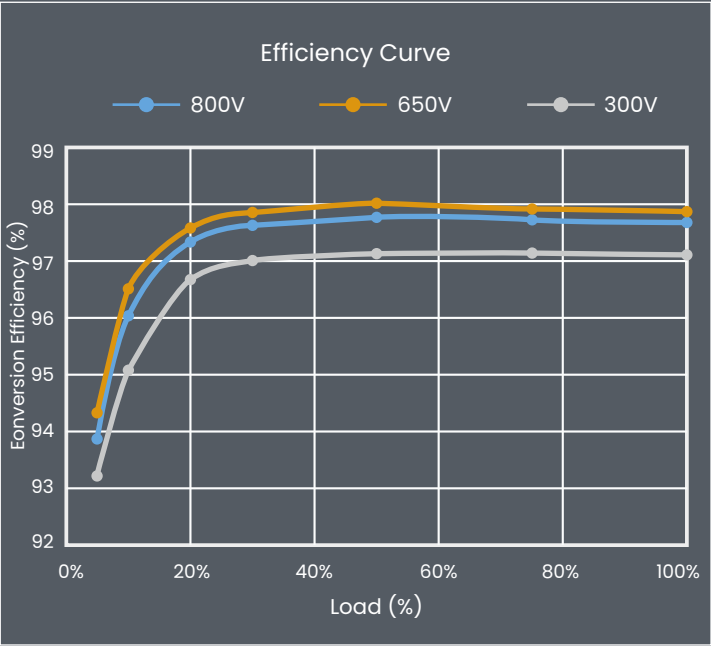
0 INJECTION SOLUTION



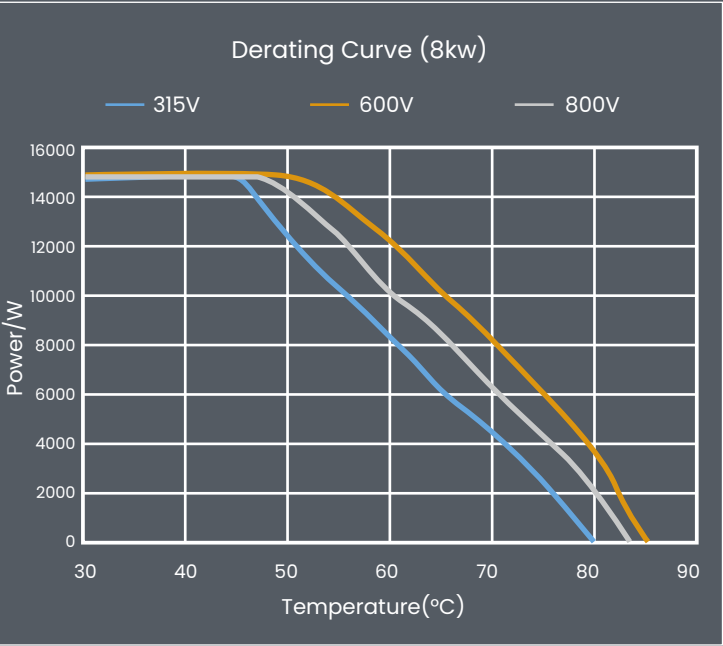
PARALLEL & GENERATOR INTEGRATED SOLUTION



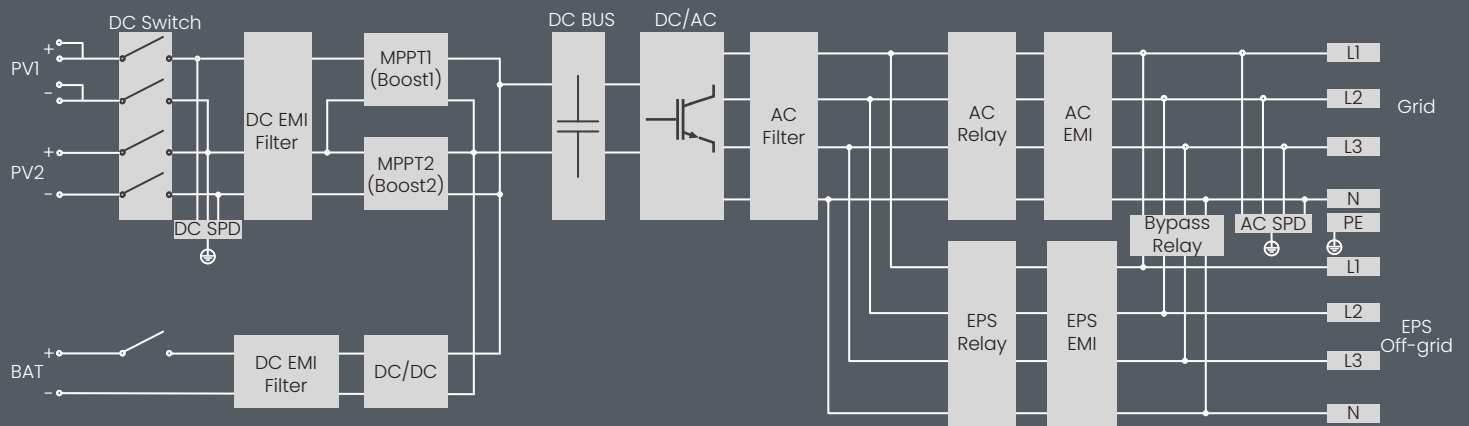
EFFICIENCY CURVE



DERATING CURVE



CIRCUIT DIAGRAM



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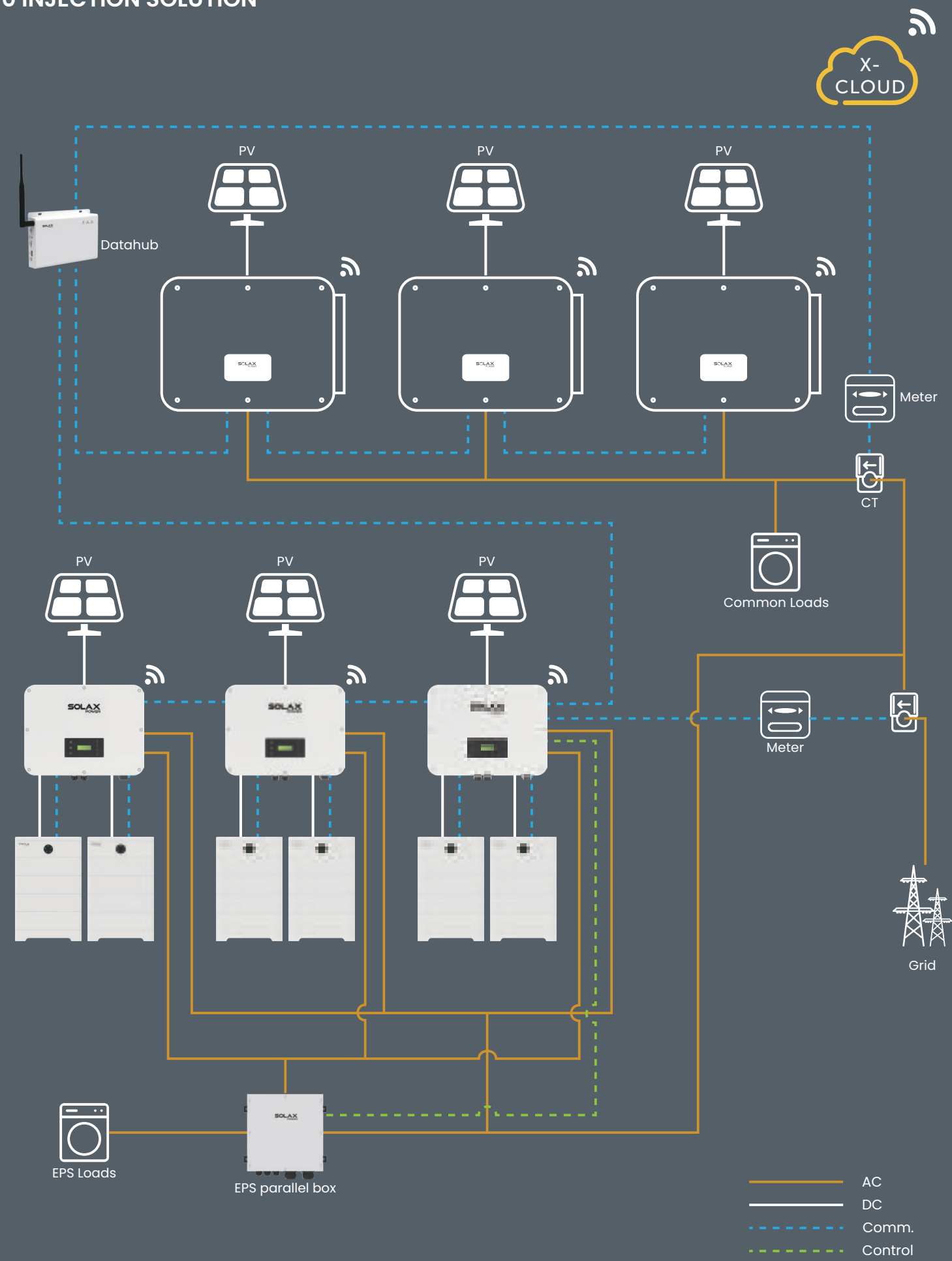
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V1.5. Information may be subject to modify without notice.
650.00039.00

0 INJECTION SOLUTION



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V1.2. Information may be subject to modify without notice.
650.00038.00



SOLAX X3-ULTRA



Features

Economic

- Maximum 200% oversize and 200% PV input power
- Maximum 36A input current per MPPT, support high power solar panel

Intelligent

- AI ready, forecasting solar generation and home consumption, smart energy management strategy
- VPP ready, SolaX cloud supports resource aggregator (2030.5, OpenADR)
- Support smart scene function, intelligent loads management(e.g., Heat pump, EV charger)
- Micro-grid ready, supporting a variety of scenarios, both on-grid and off-grid, balancing power between PCS and Hybrid in real time
- Support 7×24h scheduling mode
- Support Wireless meter solution
- Dual independent battery ports are ready to expand battery capacity free

Robust

- Robust back-up ability, switch over time <10ms, up to 200% EPS output for 10s, support half-wave loads
- Stronger, EPS operation without battery

Safe

- IP65 protection level
- AC&DC SPD type II, always guarding the inverter
- AFCI optional



X3-ULTRA (THREE PHASE)

	X3-ULT-15K	X3-ULT-19.9K	X3-ULT-20K	X3-ULT-25K	X3-ULT-30K
INPUT PV					
Max.recommended PV array power [Wp]	30000	40000	40000	50000	60000
Max.input DC power [W]	30000	40000	40000	50000	60000
Max.DC voltage [V]			1000		
Nominal DC operating voltage [V]			600		
No. of MPP trackers / Strings per MPP tracker	2 (2 / 2)	2 (2 / 2)	2 (2 / 2)	3 (2 / 2 / 2)	3 (2 / 2 / 2)
Max. input current (input PV1 / input PV2 / input PV3) [A] ^①	PV1: 36 / PV2: 36	PV1: 36 / PV2: 36	PV1: 36 / PV2: 36	PV1: 36 / PV2: 36 / PV3: 36	PV1: 36 / PV2: 36 / PV3: 36
Max. short circuit current (input PV1 / input PV2 / input PV3) [A]	PV1: 45 / PV2: 45	PV1: 45 / PV2: 45	PV1: 45 / PV2: 45	PV1: 45 / PV2: 45 / PV3: 45	PV1: 45 / PV2: 45 / PV3: 45
MPPT voltage range [V]	160 ~ 950	160 ~ 950	160 ~ 950	160 ~ 950	160 ~ 950
Start output voltage [V]	200	200	200	200	200
OUTPUT AC(On-Grid)					
Nominal AC power [VA]	15000 (AS 4777 14999)	19999	20000	25000	30000 (AS 4777 29999)
Max. apparent AC power [VA]	16500 (AS 4777 14999)	19999	22000	27500	30000 (AS 4777 29999)
Rated grid voltage (AC voltage range) [V]			3P4W, 400 / 230		
Rated grid frequency [Hz]			50 / 60		
Nominal AC current [A]	21.8	29.0	29.0	36.3	43.5
Max. AC current [A]	24.0	29.0	31.9	39.9	43.5
Displacement power factor			1 (- 0.8 ~ 0.8)		
Total harmonic distortion (THDi, rated power) [%]			< 3		
INPUT AC					
Nominal AC power [VA]	15000	19999	20000	25000	30000
Nominal AC current [A]	21.8	29.0	29.0	36.3	43.5
Rated grid voltage (AC voltage range) [V]			3P4W, 400 / 230		
Rated grid frequency [Hz]			50 / 60		
BATTERY					
Battery type			Lithium - ion		
Battery voltage range [V]			180 ~ 800		
Max. charge / Discharge current [A]			60 (30 × 2)		
EPS OUTPUT(WITH BATTERY)					
EPS peak power [VA]			2 time of rated power, 10s		
EPS rated power [VA]	15000	19999	20000	25000	30000
EPS rated voltage [V], Frequency [Hz]			400 / 230; 50 / 60		
EPS rated current [A]	21.8	29.0	29.0	36.3	43.5
Switchover time [ms]			< 10		
Total harmonic distortion (THDv, linear Load) [%]			< 3		
POWER CONSUMPTION					
Internal consumption (night) [W]			< 5		
PROTECTION					
Anti-islanding protection			Yes		
DC reverse polarity protection			Yes		
Insulation monitoring			Yes		
Residual current monitoring			Yes		
AC overcurrent protection			Yes		
AC short-circuit protection			Yes		
AC overvoltage protection			Yes		
Over-heat protection			Yes		
Battery reverse charging from grid			Yes		
Surge protection			Type II, DC and AC		
AFCI			OPT		



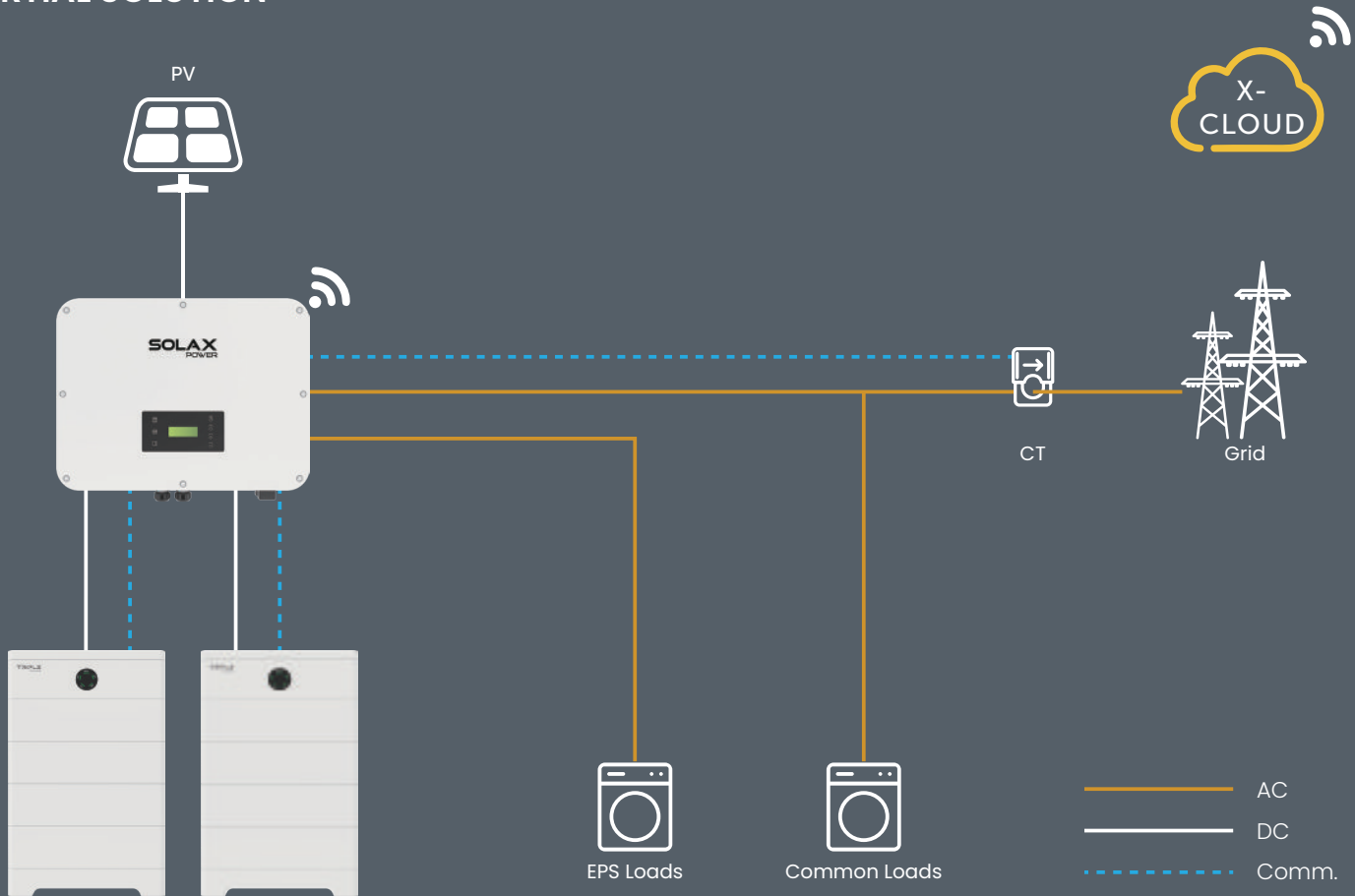
X3-ULTRA (THREE PHASE)

	X3-ULT-15K	X3-ULT-19.9K	X3-ULT-20K	X3-ULT-25K	X3-ULT-30K
EFFICIENCY					
Max. efficiency / European efficiency	98.0% / 97.7%	98.0% / 97.7%	98.0% / 97.7%	98.0% / 97.7%	98.0% / 97.7%
Rated battery charge / Discharge efficiency	98.5% / 97.0%	98.5% / 97.0%	98.5% / 97.0%	98.5% / 97.0%	98.5% / 97.0%
STANDARD					
Safety	EN / IEC62109 - 1 / - 2				
EMC	EN61000 - 6 - 1 / 2 / 3 / 4; EN61000 - 3 - 11 / 12; EN 5011; IEC 62920				
Certification	VDE4105 / G99 / AS4777 / EN50549 / CEI 0 - 21 / IEC61727 / PEA / MEA / NRS - 097 - 2 - 1 / RD1699 / TOR				
GENERAL DATA					
Protection class	IP66				
Operating temperature range [°C]	- 35 ~ 60 (Derating above + 45)				
Relative humidity [%]	0 ~ 100				
Altitude [m]	< 3000				
Storage temperature [°C]	- 40 ~ +70				
Demensions (WxHxD) [mm]	696 x 526 x 240				
Weight [kg]	47				
Cooling concept	Smart cooling				
Topology	Transformerless				
Communication	Modbus (RS485), Meter (RS485), DI x 5, DO x 2				

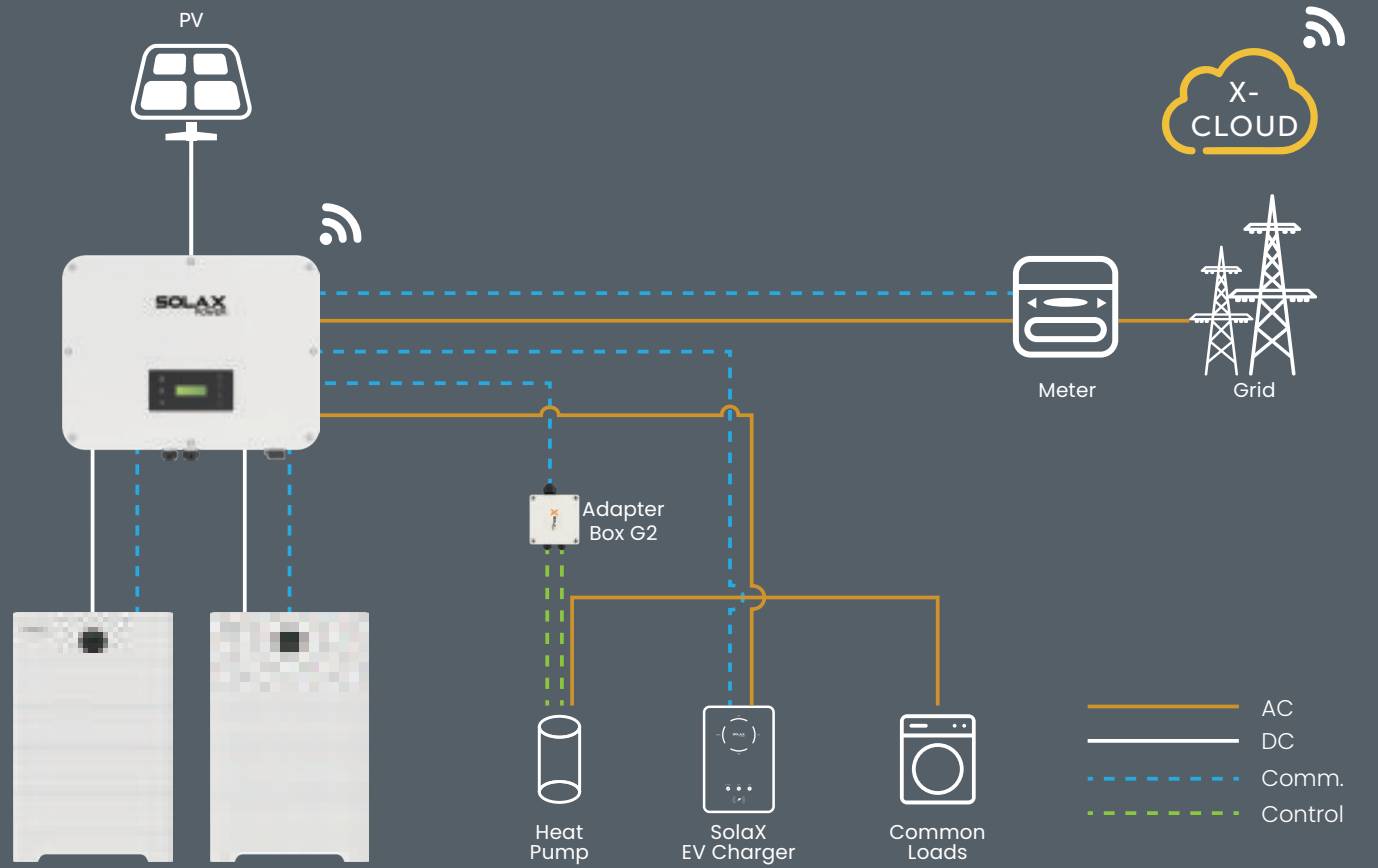
①: Input PV3 only available for 25K and 30K.

TYPICAL SCENARIO

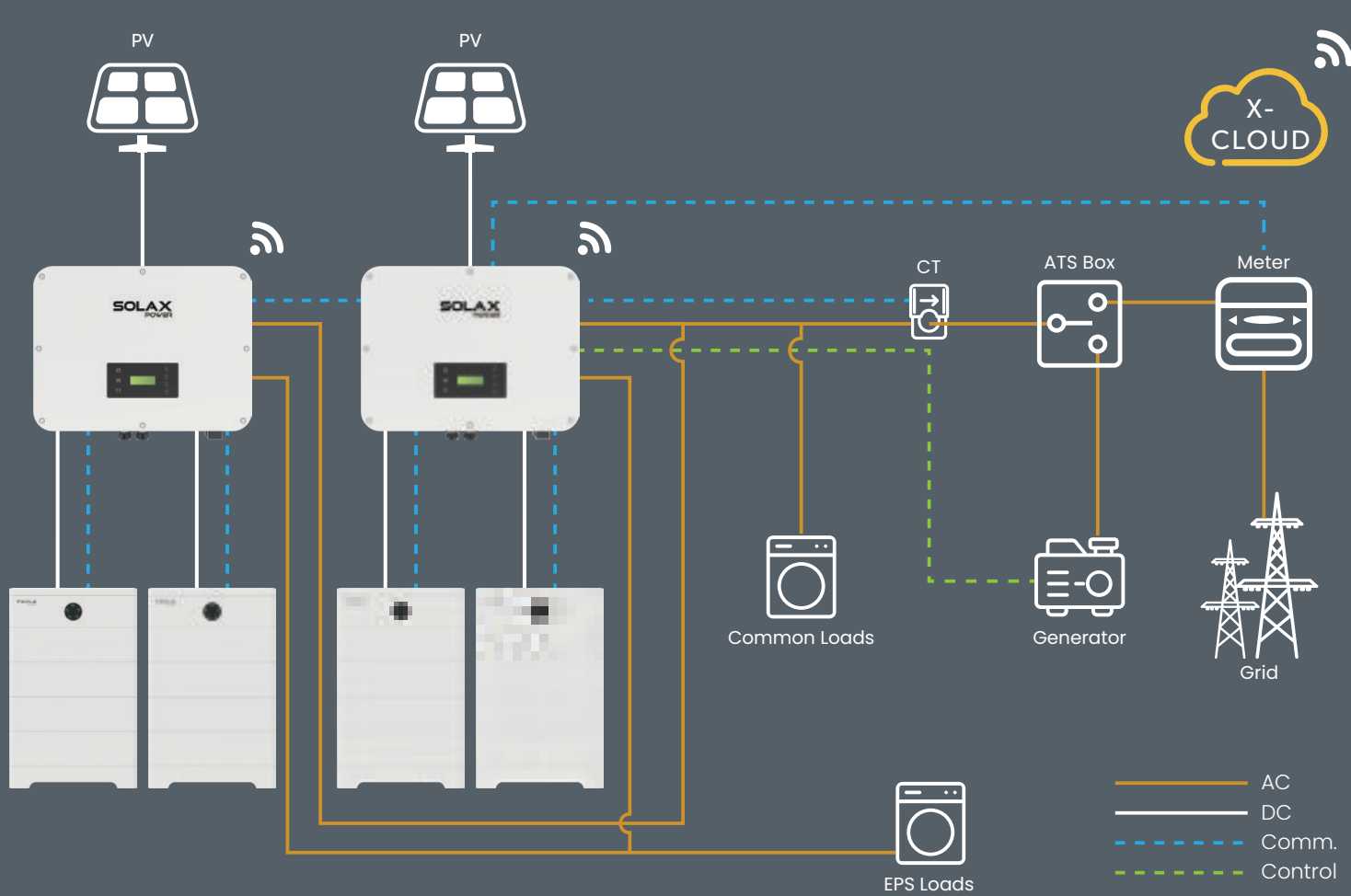
PARTIAL SOLUTION



WHOLE HOME BACKUP SOLUTION



PARALLEL & GENERATOR INTEGRATED SOLUTION



NEW FROM SOLAX

T-BAT-SYS-HV-S2.5



T-BAT-SYS-HV-S2.5

T-BAT HS5.0/T-BAT HS7.5/T-BAT HS10.0
T-BAT HS12.5/T-BAT HS15.0/T-BAT HS17.5
T-BAT HS20.0/T-BAT HS22.5/T-BAT HS25.0
T-BAT HS27.5/T-BAT HS30.0/T-BAT HS32.5

Features

HIGH-PERFORMANCE

- Max. 45A continuous charging and discharging current
- Unique battery heating technology, which is capable to work at low temperature

SAFE AND RELIABLE

- Reliable LFP battery cell
- IP65 for both indoor and outdoor installation
- Soft start protecting batteries and inverters from a sudden surge
- Cycle life > 6000 times

FLEXIBILITY

- 5.1-33.2kWh Wide capacity range
- Extendable during lifetime

EASY INSTALLATION

- Stackable modules, Easy and fast for single person installation
- Pre-wired communication cables for plug and play
- Remote diagnosis and update via inverter

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T-BAT-SYS-HV-S2.5

T-BAT HS5.0

T-BAT HS7.5

T-BAT HS10.0

T-BAT HS12.5

T-BAT HS15.0

T-BAT HS17.5

Technical Specification



2 modules



3 modules



4 modules



5 modules



6 modules



7 modules

Nominal Energy [kWh]	5.12	7.68	10.24	12.80	15.36	17.92
Usable Energy (90% DOD) ^① [kWh]	4.6	6.9	9.2	11.5	13.8	16.1
Nominal Voltage [V]	102.4	153.6	204.8	256.0	307.2	358.4
Operating Voltage Range [V]	90 - 116	135 - 174	180 - 232	225 - 290	270 - 349	315 - 406
Recommend Charge / Discharge Current ^② [A]	30					
Max. Charge / Discharge Current ^{②③} [A]	45					
Nominal Power ^④ [kW]	3.0	4.6	6.1	7.6	9.2	10.7
Max. Power ^④ [kW]	4.6	6.9	9.2	11.5	13.8	16.1
Depth of Discharge [%]	90					
Communication interface	RS485, CAN					
Dimension (L x W x H) [mm]	510 x 365 x 522	510 x 365 x 659.5	510 x 365 x 797	510 x 365 x 934.5	510 x 365 x 1072	510 x 365 x 1209.5

T-BAT HS20.0

T-BAT HS22.5

T-BAT HS25.0

T-BAT HS27.5

T-BAT HS30.0

T-BAT HS32.5

Technical Specification



8 modules



9 modules



10 modules



11 modules



12 modules



13 modules

Nominal Energy [kWh]	20.48	23.04	25.60	28.16	30.72	33.28
Usable Energy (90% DOD) ^① [kWh]	18.4	20.7	23.0	25.3	27.6	29.9
Nominal Voltage [V]	409.6	460.8	512.0	563.2	614.4	665.6
Operating Voltage Range [V]	360 - 465	405 - 522	450 - 580	495 - 636	540 - 695	585 - 750
Recommend Charge / Discharge Current ^② [A]	30					
Max. Charge / Discharge Current ^{②③} [A]	45					
Nominal Power ^④ [kW]	12.2	13.8	15.3	16.8	18.4	19.9
Max. Power ^④ [kW]	18.4	20.7	23.0	25.3	27.6	29.9
Depth of Discharge [%]	90					
Communication interface	RS485, CAN					
Dimension (L x W x H) [mm]	510 x 365 x 1347	510 x 365 x 1484.5	510 x 365 x 934.5 + 510 x 365 x 934.5	510 x 365 x 1072 + 510 x 365 x 934.5	510 x 365 x 1072 + 510 x 365 x 1072	510 x 365 x 1209.5 + 510 x 365 x 1072

T-BAT HS5.0~T-BAT HS32.5

BMS

Model	TBMS-MCS0800
Dimensions (L x W x H) [mm]	510 x 365 x 157
Weight [kg]	13

Battery Module

Battery Model	TP-HS25
Battery Type	Li-ion (LFP)
Battery Module [kWh]	2.5
Dimensions (L x W x H) [mm]	510 x 365 x 152
Weight [kg]	30
Installation Type	Stackable Level Package

SERIES BOX

Dimensions (L x W x H) [mm]	510 x 365 x 157
Weight [kg]	10

GENERAL SPECIFICATION

Installation	Floor Stand
Charge / Discharge Temperature Range (Without heating) [°C]	0 to 53 (Charge) / -20 to 53 (Discharge)
Charge / Discharge Temperature Range (With heating) [°C]	-30 to 53 (Charge / Discharge)
Max. Operating Altitude [m]	< 3000
Environment	Outdoor / Indoor (*Please refer to the user manual for installation condition)
Protection Degree	IP65
Relative Humidity [%]	4 - 100 (Condensing)

STANDARD AND CERTIFICATION

Certification	IEC62619, IEC60730, IEC62040, CE, UN38.3
---------------	--

① Test conditions: 90% DOD, 0.2C charger & discharger @+25 °C

② Max. charge / discharge current may be variant with different inverter models

③ Recommend / Max. Charging / Discharging Current* / Nominal / Max. Power*: Recommend / Max. charging/discharging current and Nominal / Max. power derating will occur related to Temperature and SOC.

BY SOLAX
TRIPLE
POWER

NEW FROM SOLAX

T-BAT-SYS-HV-S3.6



T-BAT-SYS-HV-S3.6

T-BAT HS7.2/T-BAT HS10.8/T-BAT HS14.4
T-BAT HS18.0/T-BAT HS21.6/T-BAT HS25.2
T-BAT HS28.8/T-BAT HS32.4/T-BAT HS36.0
T-BAT HS39.6/T-BAT HS43.2/T-BAT HS46.8

Features

HIGH-PERFORMANCE

- Max. 50A continuous charging and discharging current
- Unique battery heating technology, which is capable to work at low temperature

SAFE AND RELIABLE

- Reliable LFP battery cell
- IP65 for both indoor and outdoor installation
- Soft start protecting batteries and inverters from a sudden surge
- Cycle life > 6000 times

FLEXIBILITY

- 7.3-47.9kWh Wide capacity range
- Extendable during lifetime

EASY INSTALLATION

- Stackable modules, Easy and fast for single person installation
- Pre-wired communication cables for plug and play
- Remote diagnosis and update via inverter

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T-BAT-SYS-HV-S3.6

T-BAT HS7.2

T-BAT HS10.8

T-BAT HS14.4

T-BAT HS18.0

T-BAT HS21.6

T-BAT HS25.2

Technical Specification



2 modules



3 modules



4 modules



5 modules



6 modules



7 modules

Nominal Energy [kWh]	7.37	11.06	14.75	18.43	22.12	25.80
Usable Energy (90% DOD) ^① [kWh]	6.5	9.9	13.2	16.5	19.8	23.2
Nominal Voltage [V]	102.4	153.6	204.8	256	307.2	358.4
Operating Voltage Range [V]	90 - 116	135 - 174	180 - 232	225 - 290	270 - 349	315 - 406
Recommend Charge / Discharge Current ^② [A]	35					
Max. Charge / Discharge Current ^② [A]	50					
Nominal Power ^③ [kW]	3.5	5.3	7.1	8.9	10.7	12.5
Max. Power ^③ [kW]	5.1	7.6	10.2	12.8	15.3	17.9
Depth of Discharge [%]	90					
Communication interface	RS485, CAN					
Dimension (L x W x H) [mm]	510 x 365 x 522	510 x 365 x 659.5	510 x 365 x 797	510 x 365 x 934.5	510 x 365 x 1072	510 x 365 x 1209.5

T-BAT HS28.8

T-BAT HS32.4

T-BAT HS36.0

T-BAT HS39.6

T-BAT HS43.2

T-BAT HS46.8

Technical Specification



8 modules



9 modules



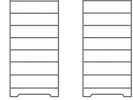
10 modules



11 modules



12 modules



13 modules

Nominal Energy [kWh]	29.49	33.18	36.86	40.55	44.24	47.92
Usable Energy (90% DOD) ^① [kWh]	26.4	29.7	33.1	36.4	39.7	43.1
Nominal Voltage [V]	409.6	460.8	512	563.2	614.4	665.6
Operating Voltage Range [V]	360 - 465	405 - 522	450 - 580	495 - 636	540 - 695	585 - 750
Recommend Charge / Discharge Current ^② [A]	35					
Max. Charge / Discharge Current ^② [A]	50					
Nominal Power ^③ [kW]	14.3	16.1	17.9	19.7	21.5	23.2
Max. Power ^③ [kW]	20.4	23.0	25.6	28.1	30.7	33.2
Depth of Discharge [%]	90					
Communication interface	RS485, CAN					
Dimension (L x W x H) [mm]	510 x 365 x 1347	510 x 365 x 1484.5	510 x 365 x 934.5 + 510 x 365 x 934.5	510 x 365 x 1072 + 510 x 365 x 934.5	510 x 365 x 1072 + 510 x 365 x 1072	510 x 365 x 1209.5 + 510 x 365 x 1072

T-BAT HS7.2~T-BAT HS46.8

BMS

Model	TBMS-MCS0800
Dimensions (L x W x H) [mm]	510 x 365 x 157
Weight [kg]	13

Battery Module

Battery Model	TP-HS36
Battery Type	Li-ion (LFP)
Battery Module [kWh]	3.6
Dimensions (L x W x H) [mm]	510 x 365 x 152
Weight [kg]	34
Installation Type	Stackable Level Package

SERIES BOX

Dimensions (L x W x H) [mm]	510 x 365 x 157
Weight [kg]	10

GENERAL SPECIFICATION

Installation	Floor stand
Charge / Discharge Temperature Range (Without heating) [°C]	0 to 53 (Charge) / -20 to 53 (Discharge)
Charge / Discharge Temperature Range (With heating) [°C]	-30 to 53 (Charge / Discharge)
Max. Operating Altitude [m]	< 3000
Environment	Outdoor / Indoor (*Please refer to the user manual for installation condition)
Protection Degree	IP65
Relative Humidity [%]	4 ~ 100 (Condensing)

STANDARD AND CERTIFICATION

Certification	IEC62619, IEC60730, IEC62040, CE, UN38.3
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① Test conditions: 90% DOD, 0.2C charger & discharger @+25 °C

② Max. charge / discharge current may be variant with different inverter models

③ Recommend / Max. Charging / Discharging Current* / Nominal / Max. Power*: Recommend / Max. charging/discharging current and Nominal / Max. power derating will occur related to Temperature and SOC.



ESS-AELIO

HYBRID C&I ESS CABINET

Solax New Commercial Energy Storage Solutions



ROBUST

- Support both on-grid and off-grid operation
- Robust back-up ability, switch over time <10ms, up to 150% EPS output for 10s
- Support unbalanced loads on three phases



SAFE

- Four-level fire safety protection
- IP66 protection for Inverter and IP54 for cabinet
- AFCI optional
- AC&DC SPD type II, always guarding the inverter
- Smart IV Curve scan for early panel diagnosis



INTELLIGENT

- AI ready, forecasting solar generation and load consumption, smart energy management strategy*
- Support smart scene function, intelligent loads management(e.g., Heat pump, EV charger)*
- VPP ready, SolaX cloud supports resource aggregator (2030.5, OpenADR)
- Support Micro-grid and a variety of scenarios
- Support 7x24h remote O&M and schedule deployment
- Support wireless meter solution



ECONOMIC

- Maximum 200% PV oversized input
- Maximum 40A input current per MPPT, support high power solar panel
- Global MPP SCAN boost solar energy harvest
- Advanced LFP battery, single cabinet with up to 200kWh, expandable to MWh

* Under development

HYBRID C&I ESS CABINET

AELIO

“INTRODUCTION”



Aelio series is a highly integrated, all-in-one, C&I Hybrid energy storage cabinet with multiple application scenarios. It has outstanding advantages such as intelligent charge and discharge management, safety and reliability, and simple operation and maintenance.

First of all, Aelio cabinet uses high-density, high-safety, and high-performance LFP batteries. There are two models with capacity of 100kWh and 200kWh. When used in a single cabinet or multiple cabinets, it can charge and discharge stably according to the set working modes at different time periods, and the large-capacity battery cell of 280Ah also reduces the initial cost of the system.

Secondly, the cabinet is equipped with a self-developed Energy Management System (EMS) that can monitor the working status and abnormal alerts of each battery cell, PCS, and fire protection system in real-time. The local data storage capability allows for data analysis and verification for up to 1 year. The advanced EMS system also has leading advantages in intelligent control of different smart operation

strategies, autonomous scheduling based on local electricity prices, and comprehensive management of photovoltaic, energy storage systems, EV charging and generators at power plant level. These features improve the overall system efficiency and shorten the investment return period.

In addition, the cabinet also has multiple safety protection measures. It has built-in protection functions such as overvoltage, overcurrent, and overtemperature. It is also equipped with fireproof materials and 4 levels of fire protection system, which can promptly detect and respond to potential fire risks, effectively control the spread of fires and reduce the risk of safety accidents.

The cabinet is suitable for various C&I PV&ESS scenarios, including peak shaving, demand response, backup mode, photovoltaic and energy storage integration, and stable load consumption curves. It also supports applications such as virtual power plants(VPP) and frequency regulation.



	AELIO-P50B100	AELIO-P50B200	AELIO-P60B100	AELIO-P60B200
DC Side				
Max. PV input power [kW]	100		120	
Max. PV input voltage [V]	1000			
Start output voltage [V]	200			
Rated input voltage [V]	650			
MPPT voltage range [V]	160 ~ 950			
No. of MPP trackers / Strings per MPP tracker	5(2 per MPPT)		6(2 per MPPT)	
Max. input current [A]	40			
Max. input short circuit current [A]	50			
AC Side				
Rated AC output power [kW]	50		60	
Rated AC output current [A]	72.2		86.6	
Max. AC output apparent power [kVA]	55		66	
Max. AC output current [A]	83.6		100.3	
Nominal grid voltage [V]	3P4W, 400/230, 380/220			
Nominal grid frequency [Hz]	50/60			
Adjustable power factor range	1 (0.8 Leading ~ 0.8 Lagging)			
THDi (Rated power) [%]	< 3			
Battery Side				
Battery type	LFP / 280Ah			
Rated battery capacity [kWh]	100	200	100	200
Rated battery voltage [V]	358.4	716.8	358.4	716.8
Battery voltage range [V]	280 ~ 408.8	560 ~ 817.6	280 ~ 408.8	560 ~ 817.6
Discharge depth [%]	90			
Rated charge/discharge current [A]	140			
Max charge/discharge current [A]	160 (80 × 2)			
General				
Dimensions(with Inverter) (W×H×D) [mm]	1310 × 2300 × 1140	2070 × 2420 × 1200	1310 × 2300 × 1140	2070 × 2420 × 1200
Dimensions (without Inverter) (W×H×D) [mm]	1020 × 2300 × 1150	1680 × 2420 × 1200	1020 × 2300 × 1150	1680 × 2420 × 1200
Weight (with Inverter) [kg]	1600	2800	1600	2800
Weight (without Inverter) [kg]	1500	2700	1500	2700
Operating ambient temperature range [°C]	-30 ~ 55			
Relative humidity (Non-condensing) [%]	0 ~ 95			
Max. operating altitude [m]	3000			
Cooling concept	Smart air cooling			
Ingress protection	Cabinet: IP55; Inverter: IP66			
Fire protection	Aerosol(Optional:Novec1230) / Water			
Topology	Non-isolated			
Standard	IIEC62619, IEC63056:2000, IEC61000, IEC62477-1, UN38.3			

HYBRID C&I ESS CABINET

Inverter



Safe

- IP66 protection level
- AC&DC SPD type II, always guarding the inverter
- AFCI optional
- Smart IV Curve scan for early panel diagnosis



Intelligent

- AI ready, forecasting solar generation and load consumption, smart energy management strategy*
- VPP ready, SolaX cloud supports resource aggregator (2030.5, OpenADR)
- Support smart scene function, intelligent loads management(e.g., Heat pump, EV charger)*
- Micro-grid ready, supporting a variety of scenarios, both on-grid and off-grid, balancing power between PCS and Hybrid in real time
- Support 7x24h scheduling mode
- Support Wireless meter solution
- Dual independent battery ports are ready to expand more capacity



Robust

- Robust back-up ability, switch over time <10ms, up to 150% EPS output for 10s
- Support off-grid operation



Economic

- Maximum 200% PV input
- Maximum 40A input current per MPPT, support high power solar panel



	X3-AELIO-49.9K	X3-AELIO-50K	X3-AELIO-60K	X3-AELIO-61K
DC Side				
Max. PV input power [kW]	100	100	120	121
Max. PV input voltage [V]	1000			
Start-up voltage [V]	200			
Nominal input voltage [V]	650			
MPPT voltage range [V]	160 - 950			
No. of MPP trackers / Strings per MPP tracker	5 (2 per MPPT)	5 (2 per MPPT)	6 (2 per MPPT)	6 (2 per MPPT)
Max. input current per MPPT [A]	40			
Max. input short circuit current per MPPT [A]	50			
AC Side				
Rated AC output power [kW]	49.9	50	60	61
Rated AC output current [A]	72.3	72.5	87.0	88.4
Max. AC output apparent power [kVA]	49.9	55	66	66
Max. AC output current [A]	83.2	83.3	100.0	101.7
Nominal AC voltage [V]	3P4W, 400/230, 380/220			
Rated AC grid frequency [Hz]	50 / 60			
Adjustable Power Factor range	1 (0.8 Leading ~ 0.8 Lagging)			
THDi (Rated power) [%]	< 3			

X3-AELIO-49.9K

X3-AELIO-50K

X3-AELIO-60K

X3-AELIO-61K

Battery Side

Battery type	Lithium - ion			
Battery voltage range [V]	180 - 820			
Max. charge / discharge current [A]	160 (80x2)			

EPS Output(with battery)

Peak apparent AC power [kVA]	49.9	55	66	66
Nominal voltage [V]; Frequency [Hz]	3P4W, 400/230V, 380/220V; 50/60Hz			
Max. output continuous current [A]	72.0	83.6	100.3	100.3
Switch time [ms]	< 10			

General

Max. efficiency [%]	98	
European weighted efficiency [%]	97.2	
Ingress protection	IP66	
Operating ambient temperature range [°C]	- 35 ~ 60	
Max. operating altitude [m]	< 3000	
Relative humidity [%]	0 ~ 100	
Typical noise emission [dB]	< 70	
Dimensions (WxHxD) [mm]	820 x 670 x 257	
Net weight [kg]	< 100	< 105
Cooling concept	Smart air cooling	
Communication interfaces	RS485, CAN(BMS), CAN, USB, DI, DO, DRM	
Display	LCD (optional)	

Power Consumption

Standby consumption (night) [W]	< 10			
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Protection

Over/under voltage protection	Yes			
DC isolation protection	Yes			
DC reverse-polarity protection	Yes			
Grid monitoring	Yes			
DC injection monitoring	Yes			
Back feed current monitoring	Yes			
Residual current detection	Yes			
Anti-islanding protection	Yes			
Over temperature protection	Yes			
SPD (DC/AC)	Type II/Type II			
Arc-fault circuit interrupter (AFCI)	OPT			

Standard

Safety	EN/IEC 62109-1/-2			
EMC	EN 61000-6-1/2/3/ 4, EN 61000-3-11/12, EN 55011, IEC 62920			
Cetification	VDE4105, G99, AS4777, EN50549, CEI 0-21, IEC61727, PEA/MEA, NRS-097-2-1, RD1699, TOR			

HYBRID C&I ESS CABINET

Pack



TB-HR140

Battery type	LFP 280Ah
Battery capacity [kWh]	14.3
Battery Configuration	1P16S
Rated cattery voltage [V]	51.2
Battery voltage range [V]	40-58.4
Weight [kg]	115
Charge / Discharge Rate	≤ 0.5C
Dimensions (WxHxD) [mm]	461 x 228 x 778
Operating temperature range [°C]	-20 ~ 53
Relative humidity (non-condensing) [%]	0 ~ 95
Max. operating altitude [m]	3000
Ingress protection	IP20
Communication to PCS	CAN



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C&I ESS CABINET

ESS-TRENE

Air cooling



PARTNER



SAFE

- Intelligent air cooling for optimal heat dissipation
- Four-level fire protection
- AC&DC type II SPD



ECONOMIC

- Advanced LFP battery quality assured
- High power density with less space
- Expandable to MWh



INTELLIGENT

- Smart energy management strategy*
- VPP ready, SolaX cloud supports resource aggregator (2030.5, OpenADR)
- Support Micro-grid and a variety of scenarios
- Support 7x24h remote O&M and schedule deployment



ROBUST

- Support on-grid and off-grid solution
- Self-developed BMS&EMS on SolaX Cloud platform
- 1 year history system data backed up on EMS
- Support Cell level balance, smart temperature check and control

* Under development



TRENE series C&I energy storage cabinet

is a highly integrated, all-in-one solution with versatile application scenarios.

TRENE series C&I energy storage cabinet is a highly integrated, all-in-one solution with versatile application scenarios. TRENE air-cooled series provides efficient, safe, and stable smart energy storage solutions.

Firstly, the cabinet adopts high-density, high-safety, and high-performance LFP cells. With a capacity of 215kWh per cabinet, it can reliably perform charging and discharging operations for single or multiple cabinets, with a lifespan of over 10 years. The large-capacity 280Ah battery cells also reduce the overall system investment cost.

Secondly, the cabinet is equipped with a self-developed Energy Management System (EMS) that can monitor the working status and abnormal alerts of each battery cell, PCS, and fire protection system in real-time. The local data storage capability allows for data analysis and verification for up to 1 year. The advanced EMS system also has leading advantages in intelligent control of different smart operation strategies,

TRENE “INTRODUCTION”

autonomous scheduling based on local electricity prices, and comprehensive management of photovoltaic, energy storage systems, EV charging and generators at power plant level. These features improve the overall system efficiency and shorten the investment return period.

Additionally, the cabinet integrates multiple safety protection measures. It has built-in protection functions such as overvoltage, overcurrent, and over-temperature, as well as fire-resistant materials and 4-level fire protection system to promptly detect and respond to potential fire risks. This effectively controls the spread of fires and reduces the risk of safety accidents.

The cabinet is suitable for various commercial and industrial scenarios, including peak shaving, demand response, backup mode, photovoltaic and energy storage integration, and stable load consumption curves. It also supports applications such as virtual power plants (VPP) and frequency regulation.

TRENE-P100B215I

AC Side	
Rated AC power [kW]	100
Rated AC current [A]	144.4
Max. AC apparent power [kVA]	110
Nominal grid voltage [V]	400 (-20% ~ +15%)
Nominal grid frequency [Hz]	50 / 60
Adjustable power factor range	0.99leading ~ 0.99lagging
THDi (Rated power) [%]	< 3
Max. efficiency [%]	98%
Battery	
Battery type	LFP 280Ah
Battery capacity [kWh]	215
Rated battery voltage [V]	768
Battery voltage range [V]	600 ~ 876
Discharge depth [%]	90
Rated charge/discharge current [A]	140
General	
Dimension (W×H×D)[mm]	1680 × 2420 × 1200
Weight [kg]	2800
Operating temprature range [℃]	-30 ~ 55
Relative humidity(Non-condensing) [%]	0 ~ 95
Altitude [m]	3000
Cooling concept	Smart air cooling
Ingress protection	IP55
Fire protection	Aerosol(Optional:Novec1230) / Water
Topology	Non-isolated
Certificates	IEC62619, IEC63056:2000, IEC61000, IEC62477-1, UN38.3, GB/T36276, GB/T34131

C&I ESS CABINET PCS



X3-TRENE-100KI

AC Side

Rated AC power [kW]	100
Rated AC current [A]	145.0
Max. AC apparent power [kVA]	110(10mins)
Nominal grid voltage [V]	3P/(N)/PE, 400/230, 380/220
Nominal grid frequency [Hz]	50 / 60
Power factor range	0.99 leading ~ 0.99 lagging
THDi (Rated power) [%]	< 3

Battery

Battery type	Lithium - ion
Battery voltage range [V]	630~900
Max. charge / discharge current [A]	140

General

Max. efficiency [%]	98
Ingress protection	IP20
Operating ambient temperature range [°C]	-25 ~ 60
Max. operating altitude [m]	3000
Relative humidity [%]	0 ~ 95
Dimensions (WxHxD) [mm]	480 × 260 × 720
Net weight [kg]	70
Cooling concept	Force air cooling
Communication interfaces	RS485/CAN/Ethernet/DI
Topology	Non-isolated

Protection

Over/under voltage protection	Yes
DC reverse-polarity protection	Yes
Residual current detection	Yes
Anti-islanding protection	Yes

C&I ESS CABINET
Pack



TB-HR140

Battery type	LFP 280Ah
Battery capacity [kWh]	14.3
Battery configuration	1P16S
Rated cattery voltage [V]	51.2
Battery voltage range [V]	40-58.4
Weight [kg]	115
Charge/Discharge Rate	≤ 0.5C
Dimensions(WxHxD) [mm]	461 x 228 x 778
Operating temperature range [°C]	-20 ~ 53
Relative humidity(non-condensing) [%]	0 ~ 95
Max. operating altitude [m]	3000
Ingress protection	IP20
Communication to PCS	CAN



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POWERING A GREEN FUTURE



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